

Dynamic Monopsony and Human Capital

William Jungerman*

UNC, Chapel Hill

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Abstract

How does dynamic labor supply change the measurement and welfare costs of monopsony? I develop a dynamic model in which a large employer competes with smaller firms for workers who accumulate human capital. The large firm internalizes how its wages shape human capital and employment across the market. Hence, the labor supply elasticity is no longer a sufficient statistic for how far wages fall below their efficient level. I characterize the efficient allocation and show that wages equal to static marginal products implement it. To quantify the losses, I solve the model with deep learning and estimate it on French matched employer–employee data. Across two representative markets, monopsony lowers welfare by 2.6% on average. Inferring productivity from a static wage equation overstates the dominant employer’s productivity and makes wages appear further below their efficient level, yet understates the welfare cost of monopsony by 22% on average.

JEL codes: C45, D43, E24, J24, J42, L13.

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*University of North Carolina, Chapel Hill. Email: wjunger@unc.edu. I am greatly indebted to Kyle Herkenhoff, Loukas Karabarbounis, Jeremy Lise, and Joseph Mullins for their invaluable advice and support. I thank Thibaut Lamadon for a very thoughtful discussion. I also wish to thank Jacob Adenbaum, Ricardo Alves Monteiro, Marlon Azinovic-Yang, Fil Babalievsky, Pierre Bodéré, Mariacristina De Nardi, Martín García-Vázquez, Axel Gottfries, Thomas Hasenzagl, Gregor Jarosch, Mahdi Kahou, Ellen McGrattan, Lukas Nord, Sergio Ocampo Díaz, Stan Rabinovich, Michael Waugh, and many conference and seminar participants for their feedback and comments. This work is supported by a public grant overseen by the French National Research Agency (ANR) as part of the “Investissements d’Avenir” program (reference: ANR-10-EQPX-17 - Centre d’accès sécurisé aux données – CASD). Special thanks to Alexandre Lebrère and François Geerolf for help with the data as well as Horng Chern Wong for sharing scripts to harmonize occupation codes. All errors are my own.

Introduction

Since the seminal work of [Robinson \(1933\)](#), labor market power has received renewed attention, both in terms of theory and measurement.^{[1](#)} Much of this work relies on static wage-setting, with welfare losses arising from too little employment and the misallocation of workers across firms. However, restricting employment also has a dynamic cost: workers miss out on the human capital they would have accumulated on the job, and once out of work or at a rival firm, they are costly to hire back. This changes employers' incentives as well: a large employer's wage choices affect the future distribution of workers across skills, employers, and non-employment, and hence future labor supply. How does dynamic labor supply, shaped by on-the-job learning and costly job changes, change the welfare costs of monopsony? And how much would we mismeasure those costs by interpreting wages through a static model?

I answer these questions by studying a dynamic model of imperfect competition in labor markets which I then quantify using French administrative data. I make five contributions. First, I document novel features of French local labor markets. I find that most markets are concentrated and, in particular, are composed of one large leading firm which is significantly larger than the other firms. Second, informed by this empirical evidence, I propose a model of a labor market in which the leading firm competes for workers with a finite number of homogeneous firms. This leading firm is non-atomistic and large enough to understand the effect its wage-setting has on the aggregate distribution of human capital. Third, I characterize the efficient allocation and show that wages equal to static marginal products decentralize it, even though workers and the dominant firm are forward-looking. In this dynamic setting, the labor supply elasticity is no longer a sufficient statistic for the wage markdown. Fourth, I solve the recursive equilibrium with deep learning, training the leading firm's value function together with its gradient to capture the dynamic incentives that determine wages. Finally, I use the estimated model to quantify the welfare losses from monopsony in French labor markets and show how inferring productivity from a static wage equation misstates productivity, markdowns, and welfare losses. Monopsony lowers welfare by 1.6% in the bottom market and 3.5% in the top, and a static interpretation of wages overstates markdowns yet understates the welfare loss

¹ See, among others, [Manning \(2011\)](#), [Card, Cardoso, Heining, and Kline \(2018\)](#), [Dube, Giuliano, and Leonard \(2019\)](#), [Lamadon, Mogstad, and Setzler \(2022\)](#), [Berger, Herkenhoff, and Mongey \(2022\)](#), and [Yeh, Macaluso, and Hershbein \(2022\)](#).

by up to 40%.

Measurement. I use rich employee-employer administrative data from France on the near-universe of workers to document several salient features of local labor markets. Defining a market as the intersection of a 4-digit occupation code and a commuting zone, I find that the majority of markets in France are very concentrated. Moreover, these markets employ a significant portion of the labor force. I also document that most markets are composed of one large *leading* firm, defined in terms of its employment share, which is significantly larger than all the other firms in the market. Finally, I examine the competitive structure of markets and find that wages at other firms barely respond to the leader's: a 1% increase in the leader's wages is associated with a 0.1% increase at other firms, consistent with modeling the other firms as atomistic.

Theory. Motivated by this empirical evidence, I propose a novel theory of imperfect competition in the labor market. I model segmented markets composed of a non-atomistic *dominant* firm and a finite number of homogeneous *peripheral* firms. Workers are heterogeneous in their general human capital which they accumulate on-the-job. Firms compete for workers by posting a menu of wages for each human capital level. There are four sources of monopsony power: 1. workers have idiosyncratic preferences for the different firms (and non-employment); 2. workers derive utility from firm-specific non-wage amenities; 3. workers are subject to switching costs whenever they transition from one firm to another or out of non-employment; and 4. the market is granular, i.e. there are a finite number of firms.

I assume the peripheral firms behave as if they are atomistic, effectively making them *static* monopsonists. They can exploit the market power due to the four sources mentioned above but do not internalize the dynamic effects when setting wages. In contrast, the dominant firm is a *dynamic* monopsonist: it is non-atomistic and internalizes how its wage choices change the future distribution of workers across employers and human capital levels. The key tradeoff is between current and future profits. Market power lets it lower wages and still keep some workers, which raises current profits. But some workers who would have stayed and accumulated skills instead move to a rival firm or into non-employment and are costly to bring back, which lowers future profits.

In a static monopsony model, the labor supply elasticity determines the wage as a share of marginal product. This relationship makes the markdown a central object in the measurement of labor market power and motivates substantial empirical work devoted to estimating it. For the dominant firm, wages also reflect how hiring today affects future profits, so the labor supply elasticity alone no longer determines the markdown. To measure the welfare costs of monopsony, I solve the social planner’s problem and show that wages equal to the *static* marginal product of labor decentralize the efficient allocation. The markdown still measures how far wages fall below their efficient level, but does not by itself capture the welfare costs of monopsony.

Computation. The main computational difficulty in solving for an equilibrium of this model is the curse of dimensionality: the aggregate distribution of workers across human capital, employers, and non-employment is a state variable. To address this, I develop a global solution method based on deep learning. I approximate worker and firm value functions with neural networks and solve their Bellman equations jointly by fitted value iteration. Since wages depend on the gradient of the dominant firm’s continuation value, I train its network on the gradient of the value in addition to its level. The recursive solution makes it possible to estimate the model at the observed distribution and compare equilibrium and efficient transition paths without assuming that the observed economy is stationary. I verify the solution independently using polynomial projection.

Results. I estimate the model separately for two different representative markets, each containing half of the workers in France. Markets are split according to the employment share of the leader: the *bottom* and *top* representative markets correspond to markets where the leader’s share is below and above the median, respectively.

The dominant firm values new workers for their contribution to future profits and therefore pays a larger share of current marginal product than its labor supply elasticity would imply. This effect is larger in the top market which is more concentrated. I find that peripheral firms pay 70 to 74% of marginal product in the bottom market and 66 to 72% in the top, compared with 88 to 91% and 82 to 85% at the dominant firm. Less-skilled workers receive the smallest share at both types of firms and in both markets.

Next, I quantify the welfare costs of monopsony. I find that welfare is 1.6% lower

in the bottom market and 3.5% lower in the top than it would be if all firms paid workers their marginal product. The efficient allocation cuts non-employment by a third in the bottom market and by half in the top, and reallocates employment toward peripheral firms. Peripheral firms initially pay a smaller share of marginal product, so eliminating wage-setting power raises their wages proportionately more and draws workers from both the dominant firm and non-employment. Monopsony also reduces opportunities to accumulate human capital: in the long-run comparison, annual learning per worker is 8.3% below its efficient level in the bottom market and 11.9% below in the top.

Finally, I ask how much a static interpretation of these wages would misstate the welfare costs of monopsony. The static wage equation attributes the future value of hiring to current productivity, overstating the dominant firm’s productivity by 4.6% in the bottom market and 11.2% in the top. This means that the share of marginal product the dominant firm pays is understated by about 4 percentage points in the bottom market and 8 in the top. Yet the inferred welfare losses are smaller, by 4% and 40%, respectively, or 22% on average across the two markets, holding worker dynamics fixed. Overestimating the dominant firm’s productivity makes the gap between its wage and marginal product look larger. It also understates how much welfare would rise if workers moved from the dominant firm to peripheral firms.

This productivity mismeasurement also changes the implied efficient allocation. In the top market, it reverses the ranking of employers: the dominant firm appears more productive than peripheral firms, while the full model’s estimate puts it slightly below them. This calls for too little reallocation toward peripheral firms. Evaluated at the full model’s productivity, implementing the implied allocation would forgo 21% of the available welfare gain. In the bottom market, the smaller productivity error leaves the dominant firm well below peripheral firms, and the implied efficient allocation barely changes.

Literature. This paper contributes to several literatures. The most closely related work is a growing literature that models imperfect competition in the labor market (Burdett and Mortensen 1998; Manning 2003; Card, Cardoso, Heining, and Kline 2018; Kroft, Luo, Mogstad, and Setzler 2025; Lamadon, Mogstad, and Setzler 2022; Berger, Herkenhoff, and Mongey 2022; Engbom and Moser 2022; Gottfries and Jarosch forthcoming; Jarosch, Nimczik, and Sorkin 2024; Volpe 2024; and Azkarate-

Askasua and Zerecero 2025, amongst many others). I build on these frameworks in two main ways. First, I introduce endogenous general human capital accumulation, so monopsony power affects workers’ future skills as well as their current employment. Second, the dominant firm is forward-looking and internalizes how its wages change the future distribution of workers across skills, employers, and non-employment.² These features alter both the measurement and welfare implications of monopsony. The dominant firm’s wage reflects the future consequences of hiring, so the labor supply elasticity no longer determines its markdown. Subsequent work by Agostinelli et al. (2025) also finds that the contemporaneous labor supply elasticity need not be a sufficient statistic for the wage relative to the marginal product of labor. The papers differ in how wages are set and what firms internalize. Their firms commit to wage-tenure contracts and take the market-wide worker distribution as given in their hiring problem, while here the dominant firm posts wages without commitment and internalizes how its choices change the future distribution. Moreover, their efficiency analysis examines employment terminations, whereas I characterize the market-wide efficient allocation and quantify welfare losses relative to it.

Search and matching models in which workers differ in their human capital also provide a framework for studying the interaction of monopsony and on-the-job learning (e.g. Barlevy 2008; Bagger, Fontaine, Postel-Vinay, and Robin 2014; Lentz and Roys 2024; Flinn, Gemici, and Laufer 2017). In contrast, my model has four sources of firms’ market power: moving costs, preference heterogeneity, amenities, and a finite number of employers. Explicitly modeling a large employer connects the market structure to observed employer sizes and allows its wage choices to affect market-wide labor supply. In this regard, the paper complements Gottfries and Jarosch (forthcoming), who study dynamic monopsony with granular employers.

The methodology draws on Ericson and Pakes (1995)-style dynamic oligopoly models³ and the literature on dynamic discrete choice (e.g. Miller 1984; Wolpin 1984; Rust 1987; Hotz and Miller 1993, amongst many others).⁴ The model can be recast as a dynamic model of multi-product firms where, instead of competing in the product market for consumer demand (oligopoly), firms compete in the input market for

² Wong (2026) extends the seminal work of Manning (2006) to allow for imperfect competition in the goods market and a richer production technology, obtaining a dynamic markdown that features the marginal profit from hiring an additional worker.

³ See Doraszelski and Pakes (2007) for a survey.

⁴ See Aguirregabiria and Mira (2010) for a survey of this literature.

workers (oligopsony). Unlike models with static product demand, labor supply here is dynamic: workers’ choices depend on future employment and human capital accumulation. The paper also contributes to the growing literature using neural networks to solve general equilibrium models,⁵ training the dominant firm’s network on Bellman derivatives as well as values (Silver et al., 2014, Czarnecki et al., 2017). Finally, in estimation, I build on the literature on estimating models of dynamic discrete choice. I use a two-step procedure in the spirit of Rust (1987) and Hotz and Miller (1993): first, I estimate the supply-side parameters by minimum distance from worker transitions conditional on observed wages; second, I estimate the demand-side parameters by matching wages.

Outline. In Section 1, I describe the data and document several salient features of French labor markets. Section 2 introduces the model and Section 3 studies the planner’s problem and characterizes the efficient allocation. In Section 4, I estimate the model and assess the model fit. Section 5 quantifies the welfare costs of monopsony.

1 Data and Descriptive Evidence

In this section, I provide a brief overview of the datasets used and document several salient features of French labor markets that will guide modeling decisions.

1.1 French administrative data

I use two French administrative matched employer-employee datasets. The DADS-Postes covers the near-universe of workers in France and is used for the descriptive evidence below. The DADS-Panel follows a subsample of workers over time and allows me to estimate human capital. Both samples cover workers aged 16–65 employed in the private sector in metropolitan France. I use DADS-Postes for 2014–2015 and DADS-Panel for 2003–2016. Education and firm revenues are unobserved, and the data do not distinguish unemployment from other non-employment. Appendix A.1 describes the datasets and sample construction.

⁵ See, among others, Maliar, Maliar, and Winant 2021; Azinovic, Gaegauf, and Scheidegger 2022; Ebrahimi Kahou, Fernández-Villaverde, Perla, and Sood 2021; Kase, Melosi, and Rottner 2025; Duarte, Duarte, and Silva 2024; Fernández-Villaverde, Hurtado, and Nuño 2023; Adenbaum, Babalievsky, and Jungerman 2024a; Adenbaum, Babalievsky, and Jungerman 2024b; Batzer 2026; Han, Yang, and E 2026; Payne, Rebei, and Yang 2026.

1.2 Defining a labor market

Deciding how to set the boundaries of a labor market is a crucial choice that will affect both estimation and results. On the one hand, too narrow of a definition mechanically results in the creation of artificial concentration. It also risks separating workers who would be applying for the same jobs. On the other hand, if defined too broadly then it may lead to markets where firms do not overlap in their competition for workers. This risks pooling workers together that do not have the same jobs.

Following [Arquié and Bertin \(2022\)](#), I define a labor market in France as the intersection of a 4-digit occupation code and a commuting zone. This decision is based on several factors. Using 1- or 2-digit codes for occupation results in very different occupations being grouped together. Table [A.3](#) in Appendix [A.2](#) illustrates occupation coding in the DADS for one of the 1-digit occupations, “Intermediate Professions”, which includes the 2-digit occupation “Intermediate health and social work professions”. Within this 2-digit occupation code, there are 21 different 4-digit occupations such as “nursing executives”, “childcare workers”, “midwives”, “eyewear opticians”, and “pharmacy assistants”. Clearly, these are very different professions. Table [A.1](#) in Appendix [A.2](#) reports self-flow rates, and Table [A.2](#) reports additional statistics for both occupation codes. As expected, the self-flow rates are higher for the less granular occupation codes: 25% of workers switch 4-digit occupations from one year to the next, compared to 17% for 2-digit occupations. A commuting zone is a geographical unit defined by INSEE within which most workers both live and work.⁶ From one year to the next, roughly 7.4% of workers work in a different commuting zone, compared to 6.9% for the département (96 in metropolitan France), and 2.9% for the region (13 in metropolitan France). Summary statistics for commuting zones can be found in Table [A.2](#) in Appendix [A.2](#).

There are 384 4-digit occupations and 304 commuting zones yielding 103,300 distinct labor markets with a self-flow rate of 72%. The majority of market outflows are due to occupation switching. Table [1](#) reports standard summary statistics on markets. The average market has 52 firms, 162 workers, and an employment Herfindahl-Hirschman Index (henceforth HHI) of 2,813. The employment HHI measures concentration as the sum of firms’ squared employment shares. The average market’s HHI of 2,813 exceeds the threshold for high concentration in the 2010 Horizontal Merger

⁶I use the 2010 classification of commuting zones. Details on methodology can be found on INSEE’s website.

Guidelines (DOJ and FTC 2010). Weighting by employment gives a different picture: the average worker is in a market with 1,412 firms, 5,349 workers, and an HHI of 836.

Table 1: Market Summary Statistics

	Mean	SD	p25	Median	p75	p95
Number of Firms (unweighted)	52	309	4	12	35	180
Number of Firms (weighted)	1,412	3,988	59	203	793	7,670
Number of Workers (unweighted)	162	916	8	30	107	596
Number of Workers (weighted)	5,349	13,316	250	836	3,039	31,865
Employment HHI (unweighted)	2,813	2,896	697	1,716	3,750	10,000
Employment HHI (weighted)	836	1,434	89	296	885	3,681
Number of Markets	103,300					

Note: Unweighted statistics computed with equal weights for each market. Weighted statistics use market employment as weights.

1.3 Market Structure

I now document several features of the competitive structure of labor markets in France which I will use to inform modeling assumptions in Section 2.

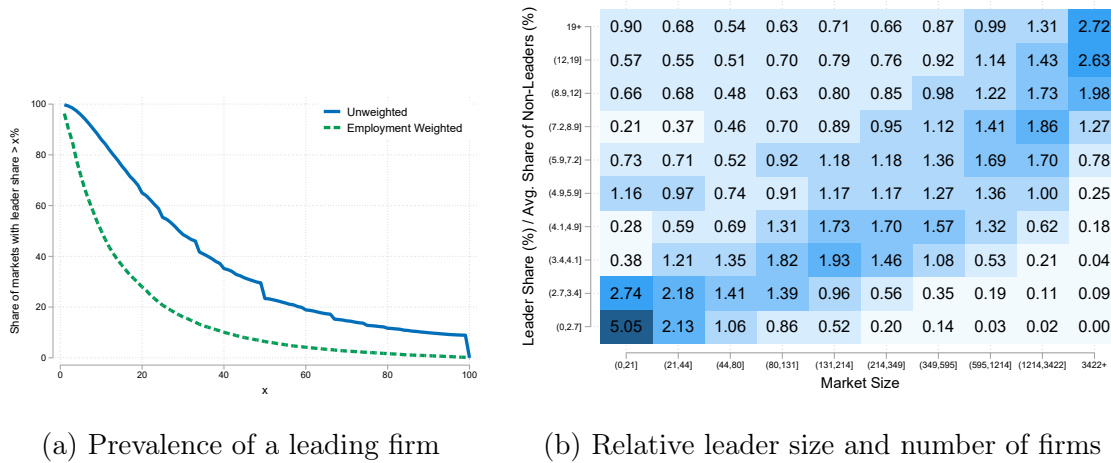
Most markets have a leader. I define the leading firm in a market as the firm with the highest employment share.⁷ Figure 1a looks at the distribution of the leader’s employment share across markets. In 65% of markets (blue solid line), the largest firm has an employment share bigger than 20%, and 27% of workers in France (green dashed line) are employed in these markets. This distribution has a heavy right tail: in 20% of markets, the share is larger than 60%. However, less than 10% of workers are in these markets.

Although informative, Figure 1a does not capture the whole picture. It could be that the markets with the largest leaders are simply ones with a very small number of firms. As Table 1 shows, these markets also tend to employ fewer workers. I investigate this further by looking at the joint distribution of the market size (number of firms in a market) and a measure of how big the leader is relative to the other firms.

⁷ In Appendix A.4, I show that the leader is persistent.

In Figure 1b, I report the employment-weighted heatmap between two variables: 1. the ratio of the leader's employment share to the mean employment share of the non-leader firms, and 2. the market size. Each number corresponds to the percentage of workers in France that are employed in these types of markets. The top left corner corresponds to the most concentrated markets: there are relatively few firms and the leader is significantly bigger than the other firms. Approximately 2% of workers are employed in these markets. Markets in the bottom left corner employ a larger fraction of workers. These are markets where there are fewer firms and the size of the leader relative to other firms is smaller. Nonetheless, these are still concentrated markets: consider for example the 2.18% of workers employed in markets where the leader is 2.7-3.4 times bigger than the other firms and there are 21-44 firms in total. Assuming the non-leader firms are of the same size and using the conservative bounds implies the leader and non-leader shares are approximately 6% and 2%, respectively. Lastly, perhaps the most striking feature of this heatmap is that conditional on the market having a very large number of firms, the gap between the leader and non-leader firms is also very large, i.e. there is no mass in the bottom right corner.

Figure 1: Market Leadership and Market Size



Note: Panel (a) plots the share of markets (solid blue) and workers (dashed green) in markets where the largest firm employs more than $x\%$ of workers. Panel (b) reports the joint distribution of the leader's employment share relative to the mean non-leader share and the number of firms. Both variables are split into employment-weighted deciles, so each row and column contains 10% of workers. Computed in DADS-Postes 2015.

The leader is much bigger than the other firms. As the y-axis of Figure 1b suggests, the vast majority of workers in France are in markets where the largest firm employs considerably more workers than the other firms in the market: 90% of workers are in markets where this ratio is larger than 2.7. In Table A.4 in Appendix A.3, I report summary statistics for several different measures to corroborate this finding. The average worker is in a market where the leader’s share is 16%, which is three times larger than the employment share of the second largest firm and ten times larger than the mean employment share of non-leader firms. These moments are driven by a heavy right tail: the median worker is in a market where the leader employs 10% of workers, which is 6 times more than the average non-leader firm and 50% more than the second biggest firm. The corresponding unweighted numbers are much larger, providing further evidence that the most concentrated markets do not employ the most workers.

Strategic Interactions. Finally, I examine the extent to which the leader and non-leader firms compete. Absent firm balance sheet data, I look at changes in wages. I estimate the following market-level specification,

$$\Delta w_{i,t}^N = \beta_0 + \beta_1 \Delta w_{i,t}^L + \epsilon_{i,t} \quad (1)$$

where i indexes the market, t is a year, and $\Delta w_{i,t}^j$ is the mean hourly wage growth, in percents, for workers at firm type $j \in \{\text{Leader}, \text{Non-Leader}\}$ from $t - 1$ to t . The leader in a market is defined as the firm with the largest employment share in $t - 1$. Figure 2 plots changes in the mean non-leader hourly wage as a function of changes in the mean leader hourly wage. The dashed green line is the regression line. The slope is $\hat{\beta}_1 \approx 0.1$ (column (1) of Table 2): a 1% increase in mean wages offered by the dominant firm is associated with a 0.1% increase in mean wages offered by the non-leader firms. Weighting by employment (column (2)) and adding fixed effects for both occupation and commuting zone (column (3)) does not significantly alter the result.

Taking stock. Defining labor markets as the intersection of a 4-digit occupation code and a commuting zone, I find that many markets in France are concentrated. Moreover, these concentrated markets employ a significant fraction of workers. I also show that markets are composed of a large leading firm that is much bigger than other

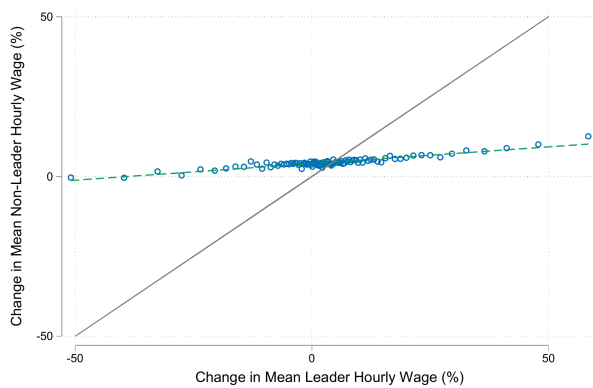


Figure 2: Wage Response of Non-Leader Firms

	Mean Non-Leader Wage		
	(1)	(2)	(3)
Mean Leader Wage	0.104 (0.005)	0.158 (0.050)	0.189 (0.050)
Fixed Effects	N	N	Y
Weighted	N	Y	Y
R ²	0.0077	0.0103	0.2595
N	96,185	96,185	96,185

Table 2: Estimates of Equation (1)

Note: Computed in DADS-Postes 2014-2015 for markets with more than one firm. **Figure.** This is a binned scatter plot with 100 bins: for each percentile along the x-axis, the mean value of the y-variable is plotted. The dashed green line is the regression line corresponding to column (1) in the table and the solid gray line is the 45-degree line. **Table.** I exclude markets for which wage growth falls outside 3 SDs of the mean. Fixed effects are for 4-digit occupation and commuting zone.

firms in the market, and that this leadership is persistent. Finally, I find suggestive evidence that the leader does not compete with the non-leader firms. These features of the data highlight the need for a model in which the largest firm knows it is big and fully internalizes how it affects market-wide outcomes.

2 Model

In this section, I present a model of a monopsonistic labor market in which workers accumulate general human capital over time. The market is composed of one large non-atomistic firm that competes with a finite number of homogeneous firms for workers. Importantly, both workers and firms make dynamic choices.

2.1 Setup

Environment. Time is discrete and the horizon is infinite, $t = 1, 2, \dots$. There is no aggregate risk.

Agents. The economy is populated by a continuum of workers with measure 1. Workers are heterogeneous in general human capital h which takes values in the discrete set $\mathcal{H} := \{h_1, h_2, \dots, h_H\}$ where $h_1 < h_2 < \dots < h_H$ and $H \in \mathbb{N}$. Workers are either non-employed or employed at a firm.⁸ At rate $\mu \in (0, 1)$, workers exogenously exit the market. Exiting workers are replaced by an equal mass of new workers, who arrive employed with a skill distribution set so that the observed distribution of human capital is stationary (see Section 4.2). Moreover, employed workers with human capital h exogenously separate from their employer at rate $\delta^h \in (0, 1)$ and become non-employed. There are two types of firms:

1. a finite number $M \in \mathbb{N}$ of homogeneous *peripheral* firms with common productivity p_P normalized to 1,⁹ and
2. a *dominant* firm with productivity p_D .

A key model ingredient is that the dominant firm is non-atomistic, i.e. it internalizes the effect its choices have on the aggregate (market-wide) state. The set of peripheral firms is denoted $\mathcal{P} := \{y_1, y_2, \dots, y_M\}$. I assume the peripheral firms behave as if they are atomistic: although there are finitely many of them, they do not internalize their effect on the aggregate distribution. This behavioral assumption is both motivated by the empirical evidence presented in Section 1 as well as tractability concerns which I discuss in Section 2.8. Note this is not as restrictive as assuming the peripheral firms behave competitively as they will still offer workers a wage below their marginal product. However, this wage (and markdown) is static and does not take into account any of the dynamics arising from the human capital accumulation of workers. I discuss this assumption more in Sections 2.5 and 2.6 when I present the firm values.

Preferences. Both workers and firms are risk neutral and discount the future with common discount factor $\beta \in (0, 1)$. Firms compete for workers by posting wages for every human capital level to maximize the present discounted value of profits. I denote these choices $w_i := (w_i^1, \dots, w_i^H)$ where $i \in \{D, P\}$ for *dominant* and *peripheral* indexes the firm type.

⁸ To be consistent with the data, I do not distinguish between non-employment and unemployment.

⁹ I deliberately call these firms peripheral rather than a competitive fringe because the usual convention is that a fringe takes wages as given, whereas the peripheral firms here do not. They are monopsonists able to price below marginal product. What they lack relative to the dominant firm is only the dynamic, non-atomistic component of wage setting, which I discuss in Sections 2.5 and 2.6.

At each period t , a worker observes their state and chooses a destination $d_t \in \{D, y_1, \dots, y_M, N\}$, where N is non-employment, to maximize the sum of present and expected future utilities. As is standard in Markov decision models, the worker's optimal decision rule is the same in two different periods if the worker's states are equal. I therefore omit time subscripts going forward. In addition to their human capital h , workers are also heterogeneous in their time-varying preferences for firms and non-employment, denoted by $\epsilon := (\epsilon_D, \epsilon_{y_1}, \epsilon_{y_2}, \dots, \epsilon_{y_M}, \epsilon_N)$. These idiosyncratic preferences reflect working conditions and commuting costs, amongst many other things, and are unobservable to the researcher. I assume that ϵ are drawn *i.i.d* from a Type 1 Extreme Value distribution every period. For every alternative $d \in \{D, y_1, \dots, y_M, N\}$, the CDF of ϵ_d is $G(\epsilon_d) = \exp(-\exp(-\epsilon_d/\sigma))$, where I normalize the location to 0 and $\sigma > 0$ is the scale parameter.¹⁰ In addition to the taste shocks, workers derive utility from non-wage amenities at both types of firms. I normalize the amenities at the peripheral firms to 0 and denote by $a_D := (a_D^1, a_D^2, \dots, a_D^H)$ the vector of amenity values workers receive at the dominant firm. Finally, workers are subject to moving costs that depend on the transition they make. A worker pays $\phi^{EE} > 0$ to move between two peripheral firms, $\phi^{PD} > 0$ to move from a peripheral firm to the dominant firm, $\phi^{DP} > 0$ to move from the dominant firm to a peripheral firm, and $\phi^{NE} > 0$ to move from non-employment to employment.¹¹ I assume utility is linear, time separable, and that the unobservable component ϵ_d enters additively into worker utility.

Technology. Firms produce a homogeneous final good with a price normalized to 1. The output produced by a firm with productivity $p_i \in \{p_D, 1\}$ and endogenous workforce $m_i := (m_i^1, m_i^2, \dots, m_i^H)$ is $F_i(m_i) := p_i \sum_{h \in \mathcal{H}} m_i^h \times h$ where m_i^h is the number of workers with human capital h hired by firm type i . I do not restrict p_D relative to the productivity of the peripheral firms: the dominant firm's position may derive from its amenities, from the moving costs workers face, or from its size rather than from a productivity advantage.¹² A non-employed worker of type h produces

¹⁰ The scale parameter governs the variance of the shocks $\pi^2 \sigma^2 / 6$.

¹¹ If there were no cost to moving from non-employment to employment, then workers could go to non-employment for one period before going to employment to avoid the employer-to-employer switching costs.

¹² This market structure is standard in dynamic oligopoly (Benkard, Jeziorski, and Weintraub, 2015) and nests a range of market structures. For example, perfect competition is reached when $p_D \rightarrow 1$, $\phi \rightarrow 0$, $a_D \rightarrow 0$, $\sigma \rightarrow 0$, and $M \rightarrow \infty$.

b^h , which can be thought of as home production.

Human Capital. A worker's human capital evolves according to their employment status. An employed worker with human capital $h = h_k$ moves up one step in the human capital ladder with probability g_E :

$$h' = \begin{cases} h_k & \text{w/ prob. } 1 - g_E \\ \min\{h_{k+1}, h_H\} & \text{w/ prob. } g_E. \end{cases} \quad (2)$$

A non-employed worker's human capital does not change hence non-employment is costly because of foregone accumulation. In the value functions below, I will use $\mathbb{E}_{h'|h,d}$ to denote the human capital dynamics of a worker with current human capital h at destination d .

2.2 Timing and Labor Market Flows

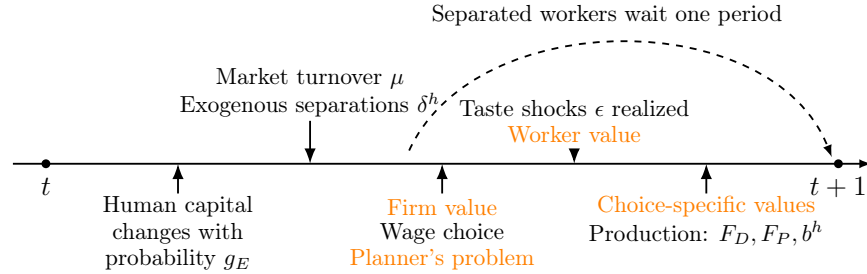


Figure 3: Within period timing of the model

Note: The timing of the value functions is in orange.

Figure 3 summarizes the timing. Human capital first evolves according to Equation (2). Market turnover then removes a fraction μ of workers, who are replaced by employed entrants with skill shares ξ^h . Among entrants of skill h , a share ξ_D^h joins the dominant firm and the rest spread evenly across peripheral firms. Next, employed workers separate at rate δ^h . Separated workers spend the period in non-employment, earning b^h , and can choose a destination only next period. This one-period wait is what makes the separation shock costly.¹³ I denote the post-turnover stocks by $\tilde{e}_D$,

¹³ Without the wait, a separated worker could re-choose immediately and, since taste shocks and

$\tilde{e}_P^y$, and $\tilde{n}$, and the post-separation stocks by e_D , e_P^y , and n . Thus, $\tilde{n}$ comprises the non-employed workers who can choose this period, while n also includes separated workers waiting until next period. Appendix B.2 gives the full laws of motion. Firms then post wages, workers observe their taste shocks and choose destinations, and production occurs. These choices determine the end-of-period stocks m_D , m_P^y , and m_N .

2.3 State Space

Firm States. In addition to its own workforce e_D and the distribution of non-employed workers n , the dominant firm only needs to keep track of the distribution of workers across *all* peripheral firms, and not the specific split into each of the M peripheral firms. This is because peripheral firms are assumed homogeneous and atomistic.¹⁴ I denote this as $s := (s^1, s^2, \dots, s^H)$ where $s^h := \sum_{y \in \mathcal{P}} e_P^{h,y}$ is total employment of workers with human capital h at the peripheral firms. Hence, the dominant firm's state is $\chi := (e_D, n, s) \in \Delta^{3H}$.

Worker States. The state of a worker is their human capital level $h \in \mathcal{H}$, their unobserved vector of preferences ϵ for each potential destination $d \in \{D, y_1, \dots, y_M, N\}$, their origin $o \in \{D, P, N\}$, and χ .

2.4 Worker Value and Mobility

Worker Value. Infinitesimal relative to the economy, workers take the aggregate laws of motion as given. A worker with state (h, ϵ, o, χ) for $o \in \{D, P, N\}$ observes the wages offered to them by the dominant and peripheral firms and decides whether to work or to be non-employed. The worker's dynamic programming problem is

amenities favor the incumbent, would typically return to it, so δ^h would be free re-optimization rather than job destruction. With the wait, displacement costs a period of earnings, a period of human capital accumulation, and requires paying ϕ^{NE} on re-entry, as in [Lucas and Prescott \(1971\)](#).

¹⁴ Another reason the split into the M peripheral firms isn't a part of the dominant firm's state is that the switching cost is the same regardless of worker origin. If the switching cost was indexed by the firm, e.g. $\phi_y^{EE} \neq \phi_{y'}^{EE}$ for two different peripheral firms $y, y' \in \mathcal{P}$, then the dominant firm would need to know which peripheral firm it is poaching from.

characterized by the standard integrated value function¹⁵

$$V(h, o, \chi) = \int \max_d [v(h, o, \chi, d) + \epsilon_d] g(d\epsilon|h, o, \chi) \quad (3)$$

where $g(\cdot)$ is the probability density function of the Type 1 Extreme Value distribution¹⁶ and $v(h, o, \chi, d)$ is the choice-specific value if alternative d is chosen. For employed destinations $d \in \{D, y_1, \dots, y_M\}$, the choice-specific value is

$$v(h, o, \chi, d) = u(h, o, \chi, d) + \beta(1 - \mu)\mathbb{E}_{h'|h, d} [\delta^{h'} U(h', \chi') + (1 - \delta^{h'}) V(h', d, \chi')]$$

where $U(h', \chi')$ is the value of being separated into non-employment given by¹⁷

$$U(h, \chi) = b^h + \beta(1 - \mu)\mathbb{E}_{h'|h, N} V(h', N, \chi'). \quad (4)$$

Current utility $u(h, o, \chi, d)$ is the sum of wages and amenities less the applicable switching cost, and in non-employment it equals home production. The choice-specific values incorporate learning and the risk of separation, including the one-period wait before separated workers can choose again. Appendix B.1 gives the full expressions and the closed-form worker value.

Conditional Choice Probabilities. The conditional choice probability $P(d|h, o, \chi)$ is the probability that a worker with states (h, o, χ) chooses destination d . To simplify notation, I write $P_{o \rightarrow d}^h := P(d|h, o, \chi)$. These choices precede next period's separation shock, whose risk the choice-specific values already include. Under Type 1 Extreme Value taste shocks, the probability of choosing destination d is

$$P_{o \rightarrow d}^h = \frac{e^{v(h, o, \chi, d)/\sigma}}{\sum_{j \in \{D, y_1, \dots, y_M, N\}} e^{v(h, o, \chi, j)/\sigma}}. \quad (5)$$

¹⁵ See Aguirregabiria and Mira (2010) for a clear survey of dynamic discrete choice models.

¹⁶ With location normalized to 0 and scale parameter $\sigma > 0$, the pdf is $g(\epsilon_d) = \exp(-\epsilon_d/\sigma - \exp(-\epsilon_d/\sigma))/\sigma$.

¹⁷ The model distinguishes two non-employment values: $U(h, \chi)$, for a worker just separated, who cannot choose this period and so receives only b^h plus a continuation; and $V(h, N, \chi)$, for a non-employed worker at the choice stage, who chooses among $\{D, P, N\}$. Equation (4) links them: a separated worker spends one period in U and becomes a chooser, $V(\cdot, N, \cdot)$, only at $t + 1$, so the one-period wait is exactly the gap between the two.

For a peripheral destination, $P_{o \rightarrow d}^h$ is the probability of choosing one particular firm: a worker with $o \in \{D, N\}$ moves to some peripheral firm with probability $MP_{o \rightarrow P}^h$, and one with $o = P$ moves to another with probability $(M - 1)P_{P \rightarrow P'}^h$.

2.5 Labor Supply Curves and Firm Value

Both types of firms $i \in \{D, P\}$ choose wages $w_i = (w_i^1, \dots, w_i^H)$ every period to maximize the present discounted value of profits. The key feature of the dominant firm's problem is that it is *non-atomistic*. It is big enough to internalize both the effect on its labor supply curve today through the choice probabilities and the effect on its future state χ' . These two effects are intertwined and introduce a fixed point problem inside of the firm's problem: to forecast χ' , the dominant firm needs to know the choice probabilities but to compute the choice probabilities, the dominant firm needs to know χ' . This subtlety arises precisely because the dominant firm is non-atomistic, a key ingredient of the model that uniquely separates it from the rest of the literature. I discuss this fixed point in more detail in Appendix C.1. Intuitively, the dominant firm's own employment is part of the aggregate state χ' , and because workers are forward-looking their choice probabilities depend on χ' . This feedback loop, which is only present because the firm is non-atomistic, is the source of the dynamic and cross-type terms in the wage equation (Section 2.6), and I derive it formally in Appendix B.6.

The assumption that the peripheral firms behave as if they are atomistic implies that they do not internalize the effect their choice of wages has on the aggregate state χ' . However, they do internalize their current labor supply curve and are able to pay workers below their marginal product of labor. Just like the dominant firm, the peripheral firms also have monopsony power due to taste shocks, non-wage amenities, switching costs, and market granularity. Even without internalizing the dynamic effects on the aggregate distribution, this still enables them to price below workers' marginal product of labor. The key difference with the dominant firm is that the wage they offer is a markdown of the *static* marginal product of labor, which I discuss more in Section 2.6. Therefore, the dominant firm is a dynamic monopsonist and the peripheral firms are static monopsonists.

Labor Supply Curves. I denote by $m_i := (m_i^1, \dots, m_i^H)$ the labor supply curve faced by firm $i \in \{D, P\}$ where m_i^h is the mass of workers with human capital h that

decide to come to firm i . The labor supply curve firms face depends on their choice of wages. An important measure of i 's market power is thus its labor supply elasticity,

$$\varepsilon_i^h(\chi, w_i) := \frac{\partial m_i^h(\chi, w_i)}{\partial w_i^h} \frac{w_i^h}{m_i^h(\chi, w_i)}, \quad (6)$$

which captures how much employment of workers with human capital h at firm i changes following a change in wages offered.

For the dominant firm, the labor supply curve for each worker type h is

$$m_D^h(\chi, w_D) = \underbrace{e_D^h P_{D \rightarrow D}^h(\chi, w_D)}_{\text{retained}} + \underbrace{s^h P_{P \rightarrow D}^h(\chi, w_D)}_{\text{poached from } P} + \underbrace{\tilde{n}^h P_{N \rightarrow D}^h(\chi, w_D)}_{\text{hired from } N}, \quad (7)$$

where the first term corresponds to retained workers, the second term to workers poached from rival peripheral firms, and the last term to workers hired from non-employment. The labor supply curves of the peripheral firms are similar,

$$\begin{aligned} m_P^h(\chi, w_P) = & \underbrace{e_D^h P_{D \rightarrow P}^h(\chi, w_P)}_{\text{poached from } D} + \underbrace{\tilde{n}^h P_{N \rightarrow P}^h(\chi, w_P)}_{\text{hired from } N} \\ & + \underbrace{e_P^{h,y} P_{P \rightarrow P}^h(\chi, w_P)}_{\text{retained}} + \underbrace{(s^h - e_P^{h,y}) P_{P' \rightarrow P}^h(\chi, w_P)}_{\text{poached from } P' \neq P}, \end{aligned} \quad (8)$$

with the main difference being an additional term that captures hiring from other peripheral firms. Notice only the retained workers do not pay the switching cost ϕ^{EE} . Hence, an inefficiency is introduced by imposing firms offer the same wages to workers regardless of origin. I discuss this more when I characterize the centralized equilibrium in Section 3.

Another implication of the dominant firm being non-atomistic is it not only internalizes the effect on its own labor supply curve but also on the labor supply curves of its rivals and the distribution of workers who choose non-employment. I define $m_N := (m_N^1, \dots, m_N^H)$ where the mass of workers with human capital h who choose non-employment is

$$m_N^h = \underbrace{e_D^h P_{D \rightarrow N}^h}_{\text{inflow from } D} + \underbrace{s^h P_{P \rightarrow N}^h}_{\text{inflow from } P} + \underbrace{\tilde{n}^h P_{N \rightarrow N}^h}_{\text{remaining at } N} + \underbrace{\delta^h \tilde{e}_D^h + \delta^h \sum_{y \in \mathcal{P}} \tilde{e}_P^{h,y}}_{\text{just separated, waiting}}. \quad (9)$$

After matches are formed, human capital changes and these labor supply curves

become χ' according to the laws of motion in Appendix B.2.

Firm Problem. We are now ready to formalize the dominant firm's problem. The dominant firm's value, $J_D(\chi)$, solves the following Bellman equation

$$J_D(\chi) = \max_{w_D \in \mathbb{R}_+^H} F_D(m_D(\chi, w_D)) - \sum_h m_D^h(\chi, w_D) \times w_D^h + \beta \mathbb{E} J_D(\chi') \quad (10)$$

subject to the labor supply curves given by Equations (7) to (9) and the law of motion for $\chi'(\chi, w_D)$ given by Equations (B.2) to (B.8). The first two terms are contemporaneous profit: output less the wage bill. The last term is the continuation value where the expectation is taken over the human capital accumulation of all the workers in the market. I relegate the first-order conditions, which I do not explicitly use to solve the model, to Appendix B.5 and a more formal discussion of the mechanics of the dominant firm being non-atomistic to Appendix B.6.

The main tradeoff in the firm's problem is between present and future profits. On the one hand, there is an incentive for the dominant firm to exploit its monopsony power and choose low wages, thus increasing today's profits. On the other hand, lowering wages affects the sorting of workers and their subsequent human capital accumulation. Some workers will opt to go to the peripheral firms, while others will choose non-employment. Both will be more expensive to re-hire in the future because of the switching costs, and in addition, the workers going to non-employment no longer accumulate human capital. Hence, by lowering wages today, the dominant firm hinders future profits.

The extent to which they exploit this market power depends on how much market power they have to begin with, as well as the learning environment in the market. I study these tradeoffs more formally and quantify their implications in Section 5.

2.6 Dynamic Markdowns

The wage equation for the dominant firm, derived in Appendix B.5, is

$$w_D^h = \eta_D^h (p_D h + d^h + c^h), \quad (11)$$

where $p_D h = \partial F_D / \partial m_D^h$ is the static marginal product, the Lerner ratio η_D^h and the dynamic term d^h are defined as

$$\eta_D^h := \frac{\varepsilon_D^h(\chi, w_D)}{\varepsilon_D^h(\chi, w_D) + 1}, \quad d^h := \beta \mathbb{E} \frac{\partial J_D(\chi')}{\partial m_D^h(\chi, w_D)},$$

and the cross-type term c^h is

$$c^h := \sum_{h' \neq h} \frac{\partial m_D^{h'}(\chi, w_D) / \partial w_D^h}{\partial m_D^h(\chi, w_D) / \partial w_D^h} (p_D h' - w_D^{h'} + d^{h'}).$$

The Lerner ratio η_D^h , which follows from the firm's labor supply elasticity as in [Lerner \(1934\)](#), applies to the dynamic marginal product of labor $p_D h + d^h + c^h$, not just to the static marginal product. The dynamic marginal product is the firm's full marginal value of attracting an additional worker of type h . The static marginal product $p_D h$ captures how much output at the dominant firm today changes as it hires an additional worker with human capital h . The dynamic term d^h captures how hiring a worker today affects tomorrow's value through changes in the aggregate distribution χ' . When the dominant firm hires a worker today, it creates future incumbents who may be more skilled and don't face switching costs when retained.

The cross-type component c^h has an intuitive interpretation. The expression $p_D h' - w_D^{h'} + d^{h'}$ is the firm's full marginal surplus from one more worker of type h' : the marginal product net of the wage plus the dynamic term. The cross-type term c^h sums these over the workers of *other* types $h' \neq h$ whose labor supply is affected by a change in w_D^h , weighted by how strong this labor supply response is. Mechanically, this shows up only because the firm is non-atomistic and large enough to move the aggregate distribution: a type- h' worker is only affected by w_D^h through its effect on the future aggregate state χ' , and hence their continuation value. The cross-response therefore exists only because the dominant firm is large enough to move χ' .

Equation (11) nests the standard static wage markdown equation. The most straightforward way to recover a static markdown is if there are no dynamics and the firm maximizes period profits (i.e. $\beta = 0$), in which case both d^h and c^h vanish. Moreover, if the firm is atomistic and does not internalize the effect its choice of wages has on its future state, then the dynamic and cross-type terms also disappear. It is

not only the fact that the firm is forward-looking that matters, but also that it is big enough to internalize its effect on the aggregate state. The wage equation for the peripheral firms, which I assume behave as if they were atomistic, is therefore

$$w_P^h = \underbrace{\frac{\varepsilon_P^h(\chi, w_P)}{\varepsilon_P^h(\chi, w_P) + 1}}_{\text{Lerner ratio}} \frac{\partial F_P(\chi, w_P)}{\partial m_P^h(\chi, w_P)}, \quad (12)$$

where I show in Corollary 1 that the Lerner ratio, $\eta_P^h := \frac{\varepsilon_P^h(\chi, w_P)}{\varepsilon_P^h(\chi, w_P) + 1}$, is equal to the ratio of the decentralized to the efficient wage. I discuss in Section 4.2 how I estimate $\varepsilon_P^h(\chi, w_P)$ internally which I then use to solve for the equilibrium of the model. Notice that the model admits a distribution of markdowns across firms, i.e. between the dominant and peripheral firms, as well as within firms, i.e. for different human capital levels.

I adopt the usual convention that the *markdown* is one minus the ratio of the wage to the static marginal product of labor, so that a higher markdown means workers receive less of their marginal product as pay. In static models, the labor supply elasticity pins down the markdown, which measures the distortion from monopsony: the efficient allocation is achieved when the firm pays the marginal product of labor, so the Lerner ratio is exactly the ratio of the wage to its efficient level. However, as pointed out by Manning (2003), it is not clear if the markdown remains the relevant measure in more general settings, which is why in Section 3 I study the centralized equilibrium of the model. In Theorem 1, I show that the *static* marginal product of labor decentralizes the efficient allocation. With dynamic wage setting, the elasticity is no longer a sufficient statistic for the markdown, the gap between the wage and its efficient level. The elasticity remains informative about wage-setting incentives, but does not by itself measure the monopsony wage distortion.

2.7 Equilibrium Concept

Definition 1 (Equilibrium). *A recursive equilibrium consists of worker values $\{V(h, o, \chi), U(h, \chi)\}$, a dominant-firm value $J_D(\chi)$, a dominant-firm policy $w_D(\chi)$, a peripheral repricing rule $w_P(\chi, w_D)$, and a law of motion for χ such that*

1. *The worker values satisfy Equations (3) and (4) given firm policies and the law of motion for χ .*

2. $J_D(\chi)$ solves Equation (10) given worker values and the law of motion for χ .
3. For every state χ and every dominant-firm wage w_D , $w_P(\chi, w_D)$ solves Equation (12) given worker values and the law of motion for χ .
4. The law of motion for χ given by Equations (7) to (9) and eqs. (B.2) to (B.8) is consistent with the worker value functions and firm policies.

The equilibrium is solved on the whole state space rather than only at the stationary distribution. The estimation and the welfare analysis rely on this: p_D is estimated from the dominant firm’s policy at the observed state, and welfare is computed along the path that starts there.

Existence and Uniqueness. The peripheral firms take the aggregate state as given. I assume that within a period they set their wages by the static Lerner rule in response to the dominant firm’s wages, and the dominant firm anticipates this response, so no best-response iteration is necessary. Standard existence arguments for pricing games require quasi-concave profits (Caplin and Nalebuff, 1991), which can fail with multi-product firms (Hanson and Martin, 1996). In Appendix B.7, I show that the dominant firm’s objective in Equation (10) need not be concave or quasi-concave. I therefore do not establish existence or uniqueness.

2.8 Solution Method

The aggregate state $\chi \in \Delta^{3H}$ is the distribution of workers across firm types and human capital levels. Even for relatively small H , the curse of dimensionality (Bellman, 1957) kicks in quickly. Suppose $H = 3$ and we solve the model on a very coarse grid with the 11 points $\{0, 0.1, \dots, 1\}$. Since $\chi \subset \Delta^{3H}$, we would not have to solve the model at all $11^9 \approx 2.4$ billion combinations, but “only” at those that sum to 1. A simple application of the stars and bars theorem shows that there are still 43,758 of them. That is feasible but coarse: half of French workers are in markets where the leader employs less than 10% (Section 1), and a grid with spacing 0.02, fine enough to resolve such shares, has about 1.9 billion points.¹⁸ In addition to the curse of

¹⁸ Tensor-product polynomial bases (Chebyshev, Hermite) face the same curse: the number of basis functions scales exponentially in the dimension of χ .

dimensionality, the dominant firm’s non-concave value (Lemma 1) is the other complication, since first-order conditions are necessary but not sufficient. I therefore solve the firm’s wage choice by multi-start numerical optimization at each state χ , rather than relying on first-order conditions or approximating the policy with a separate network.

I solve for the recursive equilibrium by fitted value function iteration, where the two value functions $V(h, o, \chi)$ and $J_D(\chi)$ are approximated with neural networks.¹⁹ Two structural features of the model make this tractable:

1. *The worker problem is closed-form.* Given the continuation values at next period’s state, the conditional choice probabilities and this period’s worker values are the standard logit expressions (Equation (B.1) and eq. (5)).
2. *Peripheral wages are a repricing rule.* At every state the peripheral firms’ wage solves $w_P^h(\chi, w_D) = \eta_P^h(\chi, w_D) \times h$. It is computed inside the within-period fixed point for every candidate dominant-firm wage, so no strategic best-response loop is needed.

Given an initialization of the networks approximating the worker and dominant firm values, I first train the worker network with gradient descent to satisfy the Bellman equations in Equations (3) and (4). I then solve the dominant firm’s wage choice with multi-start optimization and train its value network. I repeat these two steps, updating the continuation values between iterations. Because the gradient of the firm’s continuation value enters its wage-setting first-order condition, I train the firm network on the gradient of the Bellman equation as well as its level (Silver et al., 2014, Czarnecki et al., 2017). For both training steps, I mix Sobol points on the simplex with draws around the current estimate of the stationary state, placing additional weight on states near the economy’s long-run allocation of workers across firms. Appendix C describes the solution method in full.

I assess the accuracy of the solution method in two ways. First, I evaluate the Bellman equations and the gradient of the firm value along simulated equilibrium paths starting at the observed state and alternative initial distributions. I express worker Bellman residuals relative to wages and translate errors in the firm’s continuation-

¹⁹ Neural networks can approximate any continuous function on a compact domain to arbitrary accuracy (Hornik, Stinchcombe, and White 1989). See Goodfellow, Bengio, and Courville (2016) for a textbook treatment.

value gradient into more interpretable implied wage errors. As described in Section 4.1, I estimate the model for two separate markets. Across the many simulated states per market, the mean worker Bellman residual is 0.13% and 0.11% of the mean wage at the skill level in the two markets²⁰. The implied wage error of the dominant firm averages 0.24% and 0.08% of the wage. Second, I independently verify the solution by solving the model with a different method: polynomial projection (Judd, 1992, Judd, Maliar, and Maliar, 2011). This check fits low-degree polynomials on a local domain covering the relevant transition paths. During estimation, each candidate parameter vector changes those paths and the stationary state, so using this local method would require adapting and checking the approximation domain across candidate parameter values, making it infeasible for estimation. However, it can be used as a post-estimation check on the solution method once the domain is known. At the estimated parameter values, the two solutions set the leader’s wage within 1% of each other at every simulated state. Plots of the errors and more accuracy checks are reported in Appendix C.2.

3 The Centralized Equilibrium

Does allowing for human capital accumulation exacerbate or mitigate the inefficiencies from firms having monopsony power? To better understand the welfare costs of monopsony in the presence of dynamic human capital accumulation, in this section I study the centralized equilibrium of the model.

3.1 The Planner’s Problem

Consider a social planner that chooses the allocation of workers across firms and non-employment to maximize market-wide welfare. The planner chooses at the same time as firms set wages, faces the same learning, turnover, and separation processes, including the one-period wait after separation, and has the same state $\chi = (e_D, n, s)$. As in Lagakos, Mobarak, and Waugh (2023) and Waugh (2023), I express his choices as transition probabilities $\mathbb{P} \subset [0, 1]^{10 \times H}$. These determine employment $m_{D,S}$ and $m_{P,S}$, non-employment $m_{N,S}$, the masses making each transition, and the expected

²⁰ Because utility is linear in income, a Bellman residual of r is equivalent to consumption being off by r in the current period. The ratio is therefore the relative consumption error, the standard accuracy measure in the deep learning literature (Azinovic-Yang and Žemlička, 2025).

taste-shock utility $\mathcal{E}(\mathbb{P})$. Appendix B.3 gives the full setup and expressions for these objects. The planner's value solves

$$\begin{aligned}
S(\chi) = \max_{\mathbb{P} \in [0,1]^{10 \times H}} & \left\{ F(m_{D,S}(\mathbb{P})) + MF(m_{P,S}(\mathbb{P})) + \sum_h m_{N,S}^h(\mathbb{P})b(h) \right. \\
& + \mathcal{E}(\mathbb{P}) + \sum_h m_{D,S}^h(\mathbb{P})a_D^h - \phi^{DP} \sum_h m_{D \rightarrow P,S}^h - \phi^{PD} \sum_h m_{P \rightarrow D,S}^h \\
& \left. - \phi^{EE} \sum_h m_{P \rightarrow P',S}^h - \phi^{NE} \sum_h m_{N \rightarrow E,S}^h(\mathbb{P}) + \beta \mathbb{E}S(\chi') \right\} \quad (13)
\end{aligned}$$

subject to the worker flows and probability constraints in Appendix B.3 and the law of motion in Appendix B.2. The first three terms of the planner's objective are market-wide output, the fourth is the total value of the taste shocks, the fifth is the total amenity values, and the last four are the moving costs paid by movers, which differ by the type of transition. Like the dominant firm's problem, the planner's problem is dynamic.

Inefficiencies. The only inefficiency in the model is allocative. Firms cannot price discriminate according to a worker's origin: the wage offered to a worker with human capital h is the same whether the worker is retained, poached from a rival firm, or hired from non-employment, even though these workers are subject to different switching costs. A firm with market power therefore marks down the wage of all three groups, which distorts where workers go.²¹

3.2 Decentralization

I now characterize the wages that decentralize the efficient allocation of workers to firms and non-employment, which I denote $w_{D,S}^h$ and $w_{P,S}^h$.

Theorem 1 (Decentralization). *The static marginal product of labor decentralizes the efficient allocation:*

$$w_{D,S}^h = p_D \times h, \quad w_{P,S}^h = h. \quad (14)$$

The proof is in Appendix B.4. This result is reminiscent of the seminal work of Becker (1962). With constant returns to scale production, the destination choice of

²¹ As in Mongey and Waugh (2024), markets are incomplete and workers cannot insure against their idiosyncratic taste shocks. However, since utility is linear in income, this missing market does not distort choices and creates no inefficiency.

any one worker does not impact either pay or productivity of any other worker in the economy. In other words, the worker's choice does not have any spillovers to other workers. So if workers receive the full value of the output they produce while employed, then they internalize the full set of costs and benefits of their decisions. Hence offering them their *static* marginal product of labor results in the efficient allocation. Perhaps surprisingly, there is no dynamic component. The reason for this is two-fold. One contributing factor is that the worker is forward-looking and already internalizes the present discounted value of wages they can make in the future. Additionally, the worker is atomistic so does not internalize how their choice can affect the aggregate distribution.

Next, I define ν_i^h for $i \in \{D, P\}$ and $h \in \mathcal{H}$ to be the equilibrium wage divided by the wage that decentralizes the planner's allocation.

Corollary 1. ν_P^h is equal to the Lerner ratio η_P^h : $\nu_P^h := w_P^h/w_{P,S}^h = w_P^h/h$.

Corollary 1 follows immediately from Theorem 1 and is the standard result found in static models of monopsony: the Lerner ratio $\eta_P^h = \varepsilon_P^h(\chi, w_P)/(\varepsilon_P^h(\chi, w_P) + 1)$ is exactly the ratio of the wage to its efficient level, so one minus it, the markdown, measures the distortion from monopsony. In Section 4.2, I describe how I estimate $\varepsilon_P^h(\chi, w_P)$ internally which I then use to compute η_P^h and solve the model.

Corollary 2. ν_D^h need not equal the Lerner ratio η_D^h : $\nu_D^h := w_D^h/w_{D,S}^h = w_D^h/(p_D \times h)$.

Corollary 2 also follows immediately from Theorem 1. There are several takeaways. Note that $p_D = 1$ does not imply that the market is perfectly competitive. The wages the dominant firm offers w_D^h are below the worker's marginal product even if $p_D = 1$ because it still has monopsony power from the taste shocks, switching costs, amenities, and granularity of the market. Moreover, although the wages that decentralize the efficient allocation are static, ν_D^h is dynamic: w_D^h is set according to the wage equation given by Equation (11) which contains a dynamic term arising from human capital accumulation and from employer attachment due to switching costs and turnover. Lastly, ν_D^h is *not* equal to the Lerner ratio $\eta_D^h = \varepsilon_D^h(\chi, w_D)/(\varepsilon_D^h(\chi, w_D) + 1)$. Substituting in Equation (11) yields an expression for ν_D^h in terms of the Lerner ratio and the continuation and cross-type terms, $\nu_D^h = \eta_D^h[1 + (d^h + c^h)/(p_D h)]$. The difference between ν_D^h and η_D^h depends on the sum of the continuation and cross-type terms relative to static marginal product. Neither ratio alone determines the

welfare loss. As mentioned in Section 2.6, the usual static case where $\nu_D^h = \eta_D^h$ can be recovered if the dynamic and cross-type terms are zero, which can occur if $\beta = 0$, or if the dominant firm is atomistic and does not internalize its effect on the aggregate distribution χ' .

The gap between ν and η has important implications for the measurement of monopsony power. In static models and for the peripheral firms here, the labor supply elasticity pins down the markdown. With dynamic wage setting, the elasticity is no longer a sufficient statistic for the markdown: recovering ν_D^h also requires the continuation and cross-type terms relative to static marginal product and η_D^h alone does not determine this ratio. In Section 5.3, I quantify the errors in estimates of productivity, wage markdowns, and the welfare cost of monopsony that arise from ignoring dynamics.

4 Estimation

4.1 Representative Markets and Moments

The model describes a single local labor market, while the data contain tens of thousands. I therefore pool markets into two groups according to the employment share of their leading firm. The split is at the employment-weighted median, so that each group accounts for half of employment. I make this split using the DADS-Postes, then construct the estimation moments in the DADS-Panel.²² Table 3 summarizes the two representative markets, which I label “bottom” and “top”. The bottom market has thousands of firms and an employment HHI of 121, but even there the leader employs about five times as many workers as the average other firm. The top market has fewer firms, an HHI above 1,300, and a leader with about a quarter of employment.

Non-employment. Because the DADS-Panel records employment spells, I infer non-employment from gaps between jobs. I label a worker as non-employed in a year with no job if they are employed in both an earlier and a later year, and attach them to the market of their most recent job. I exclude workers with a spell of three or more

²² The DADS-Panel contains roughly one in twelve workers, so I use the more comprehensive DADS-Postes to measure firm shares and concentration. A finer split, or more skill levels, would leave small transition cells: in the bottom market, the non-employment-to-leader cells contain only about 83 workers per skill level, while disclosure requires at least ten observations.

Table 3: Representative Markets Summary Statistics

	Bottom	Top		Bottom	Top
Number of Firms	2,988	260	Leader Share (%)	4.9	26.3
Number of Workers	10,971	1,929	Avg. Non-Leader Share (%)	0.95	4.89
Employment HHI	121	1,331	Number of Markets	8,725	18,090

Note: Means weighted by market employment. Computed in DADS Postes 2015.

consecutive years of non-employment so that the sample does not include workers out of the labor force.

Human capital levels. In each market, I use *k-means* to group worker-year observations into $H = 3$ levels²³ based on experience, pooling employed and non-employed workers. In the model, human capital accumulates with employment, making experience the natural empirical counterpart. Pooling ensures that the classification is common across the dominant firm, peripheral firms, and non-employment.²⁴ I then pool workers at each level across the markets in a representative group so each market contributes in proportion to its employment. The estimation uses mean wages by skill level and employer type, $w^{\text{data}} = (w_D^{\text{data}}, w_P^{\text{data}})$, thirty annual transition rates grouped by skill in the destination year, P^{data} , and the distribution of workers across skills and employment states.

4.2 Estimation Strategy

Given firms' wages, the supply parameters alone determine worker mobility. Given the supply parameters, the dominant firm's wage setting depends only on its productivity p_D . This structure allows me to use a two-stage procedure commonly used in the dynamic discrete choice literature. I first estimate the supply parameters in partial equilibrium, independent of p_D , from worker transitions at observed wages. The human capital grid does not need to be estimated: it follows from $w_P^h = \eta_P^h h$ once the supply parameters give η_P^h . Second, I estimate p_D by targeting the dominant firm's

²³ Because the taste shocks have full support, the model places workers in every combination of skill level and employment state, so each market must have workers at the leader, at the other firms, and in non-employment.

²⁴ Clustering on wages would confound worker skills with firms' wage policies and would require imputing wages to the non-employed.

wages in the recursive equilibrium.

Externally set. The model is estimated at an annual frequency so I set $\beta = 0.95$. I also set the market exit rate to $\mu = 0.25$, the rate at which workers permanently leave their market in the DADS-Panel.²⁵ I set $b^h = \rho w_P^{h,\text{data}}$, with $\rho = 0.85$ in both markets, because transition rates barely separate ρ from the scale of taste shocks.²⁶ Because conditional choice probabilities are nonlinear in M , the number of firms M is not the average of the pooled markets. Hence I calibrate M to match the probabilities pooled across the observed distribution of firm counts (see Appendix B.8 for more details). This gives $M = 667$ for the bottom market and $M = 127$ for the top market. Given the supply estimates, I set the entrant skill mix ξ to make the observed skill distribution stationary under the law of motion at observed wages and the estimated choice probabilities.

Supply estimation. Let $\theta_S := \{\phi^{EE}, \phi^{PD}, \phi^{DP}, \phi^{NE}, \sigma, g_E, a_D^1, a_D^2, a_D^3, \delta^1, \delta^2, \delta^3\}$ denote the supply parameters. Given wages, they determine worker values and mobility without requiring a solution to the firm's problem. I estimate them by minimizing the mean absolute distance between observed and model-implied transition rates:

$$\hat{\theta}_S = \arg \min_{\theta_S} \frac{1}{30} \sum_{i=1}^{30} |P_i^{\text{data}} - P_i(\theta_S; w^{\text{data}})|$$

where wages are held fixed at their observed values. I therefore estimate θ_S in partial equilibrium, independent of p_D . Worker values are computed as if observed wages persisted in every future period, which is a departure from the full model where future wages instead depend on the evolving distribution of workers. I later check this approximation using the estimated model.²⁷

²⁵ Exit is defined to be the last year a worker is observed in a market. The employment-weighted exit and entry rates are 0.252 and 0.269 and vary little across leader-share deciles.

²⁶ As is well known for this class of models, choices reveal only relative values (Rust, 1987, Magnac and Thesmar, 2002). The transition rates change very little when σ , the four switching costs, $1 - \rho$, and the dominant firm's compensation gap all change in the same proportion. Re-estimating at $\rho = 0.80$ and 0.90 shifts markdown levels but leaves their ordering across human capital levels and firm types unchanged. Since $p_P = 1$, home production relative to the marginal product is $b^h/h = \rho \eta_P^h$, which is 0.56 to 0.63 across skill levels and markets, between the values of 0.4 in Shimer (2005) and 0.955 in Hagedorn and Manovskii (2008).

²⁷ The mean absolute error of the thirty transition rates is 0.55 and 0.52 percentage points in the first stage (bottom and top markets), 1.58 and 0.76 in the full equilibrium at the observed leader

Heuristic Discussion. While the supply-side parameters are all jointly estimated, a heuristic discussion about which moments are most informative for each of the parameters is helpful. Staying at a peripheral firm rather than moving to another identifies the switching cost ϕ^{EE} relative to the taste-shock scale σ .²⁸ Movements into and out of the dominant firm, together with the observed wage gaps, inform its amenities and the costs ϕ^{PD} and ϕ^{DP} . Transitions out of non-employment inform the entry cost ϕ^{NE} , while movements into non-employment reflect both voluntary quits and exogenous separation. Variation in these margins across human capital levels helps distinguish learning, which changes the value of future employment, from the common scale of taste shocks.

Peripheral elasticities. For each human capital level h , I estimate the peripheral firms' labor supply elasticity $\hat{\varepsilon}_P^h$ using $\hat{\theta}_S$ and the observed distribution of workers. Specifically, I compute the change in employment at a peripheral firm following a 1% unexpected decrease in the wage it offers to workers with human capital h , keeping all other wages fixed. The labor supply elasticity is then computed as the percentage change in employment. Given the estimated elasticity, I apply the Lerner formula for a static monopsonist $\hat{\eta}_P^k = \frac{\hat{\varepsilon}_P^k}{1 + \hat{\varepsilon}_P^k}$ to recover the Lerner ratio, and the peripheral wage equation $h_k = \frac{w_P^{k, \text{data}}}{\hat{\eta}_P^k}$ pins down the human capital grid.

Demand estimation. Worker transitions conditional on wages inform the dominant firm's amenities and the costs of moving into and out of it. Given these supply estimates and the human capital grid, the firm's wage schedule then informs its productivity through wage setting: I choose p_D to match the dominant firm's wage schedule:

$$\hat{p}_D = \arg \min_{p_D} \frac{1}{H} \sum_h \frac{|w_D^{*h}(\chi_{\text{data}}; p_D) - w_D^{h, \text{data}}|}{w_D^{h, \text{data}}}.$$

This requires solving the full recursive equilibrium. Evaluating the wage policy at the observed state χ_{data} , rather than at the model's stationary distribution, treats observed wages as equilibrium choices without assuming that the observed distribution

wages, and 3.40 and 1.18 at the fitted leader wages.

²⁸ To see this, note that the log odds of staying at a peripheral firm rather than moving to any other are $\ln [P_{P \rightarrow P}^h / ((M-1)P_{P \rightarrow P'}^h)] = \phi^{EE}/\sigma - \ln(M-1)$, which pins down ϕ^{EE}/σ given M .

is stationary.

4.3 Parameter Estimates

Table 4: Parameter Estimates

		Bottom	Top			Bottom	Top
Amenities at D	a_D^1	4,370 (100)	−930 (90)	Separation rates	δ^1	0.223 (0.008)	0.171 (0.008)
	a_D^2	8,470 (60)	−1,260 (30)		δ^2	0.104 (0.002)	0.071 (0.001)
	a_D^3	6,130 (80)	−1,820 (50)		δ^3	0.098 (0.003)	0.076 (0.003)
Switching cost ($P \rightarrow P'$)	ϕ^{EE}	24,150 (80)	27,940 (80)	Learning rate	g_E	0.112 (0.003)	0.081 (0.004)
	ϕ^{PD}	18,550 (130)	20,930 (90)	Home production	b^1	19,560	18,070
	ϕ^{DP}	10,020 (140)	10,380 (80)		b^2	30,230	25,910
	ϕ^{NE}	29,950 (90)	27,790 (90)		b^3	33,000	28,490
Scale of taste shocks	σ	2,630 (7)	3,490 (6)	Entrant skill mix	ξ^1	0.44	0.38
Home production fraction	ρ	0.85	0.85		ξ^2	0.36	0.37
Peripheral firms	M	667	127		ξ^3	0.19	0.25
Productivity of D	p_D	0.694	0.984				

Note: Switching costs, amenities, home production, and the taste-shock scale are in 2015 euros. Parentheses report standard errors from 300 bootstrap replications. Home production is fixed at $b^h = 0.85w_P^{h,data}$; it does not move with counterfactual wages. The entrant mix and firm count are calibrated as described in Section 4.2. The dominant firm’s shares of entrants, ξ_D^h , equal its observed employment shares at each skill level: (0.020, 0.022, 0.025) in the bottom market and (0.210, 0.231, 0.231) in the top.

Table 4 reports the estimates for both markets. The amenity estimates mirror the observed wage patterns (Table 6). In the bottom market the dominant firm pays less than peripheral firms at every skill level, but offers positive amenities worth about 4,400€ to 8,500€ per year. Including these amenities leaves its current compensation about 1,500€ to 2,200€ above peripheral compensation. In the top market the dominant firm pays a wage premium but has negative estimated amenities, leaving a net premium of about 3,700€ to 4,400€.

Switching costs are substantial and have a similar structure in both markets. Moving between peripheral firms costs about 24,000€ in the bottom market and 28,000€ in the top. Entering employment from non-employment costs about 30,000€ and 28,000€.²⁹ Moves involving the dominant firm are cheaper: joining it costs about three-quarters of the cost of moving between peripheral firms, while leaving it costs less than half. These costs summarize search, information, and other nonpecuniary

²⁹ These costs are large, between 0.6 and 1.3 times the annual peripheral wage, but smaller relative to wages than estimates of the cost of moving across sectors: several times the annual wage in [Artuç, Chaudhuri, and McLaren \(2010\)](#) and 1.4 to 2.7 times in [Dix-Carneiro \(2014\)](#).

frictions rather than cash payments. Workers are more likely to move when they receive a favorable taste shock at the destination.

The annual learning probability is 0.112 in the bottom market and 0.081 in the top, corresponding to about nine and twelve years of employment per upward skill transition below the highest level. Separation rates are highest at the lowest skill level, at 0.22 and 0.17, compared with 0.07 to 0.10 at the other levels. Low-skill workers therefore have more human capital left to accumulate but face a greater risk of an interruption to employment.

Finally, the dominant firm is less productive than peripheral firms in the bottom market and about as productive in the top: p_D is 0.694 and 0.984, relative to peripheral productivity normalized to one. Relative productivity is not the relative wage, because the two types of firms set wages differently. Combining Equations (11) and (12) gives

$$p_D = \underbrace{\frac{w_D^h}{w_P^h}}_{\text{relative wage}} \underbrace{\frac{\eta_P^h}{\eta_D^h}}_{\text{relative Lerner ratio}} - \underbrace{\frac{\eta_P^h(d^h + c^h)}{w_P^h}}_{\text{dynamic adjustment}}. \quad (15)$$

In the estimated model, the bottom market's dominant firm pays 13 to 16% less than peripheral firms, yet pays a larger share of its marginal product (relative Lerner ratios of 0.82 to 0.87), and part of its wage reflects the continuation and cross-type terms (an adjustment of 0.01 to 0.03). All three point to lower productivity. In the top market it pays 17 to 23% more, but its larger share of marginal product (relative Lerner ratios of 0.88 to 0.93) and the dynamic adjustment (0.09 to 0.11) account for the premium, leaving productivity close to parity. Size and productivity are therefore not linked here. The dominant firm is large because workers value it: in the bottom market its amenities more than offset its lower wages, and moving to it is cheaper than moving between peripheral firms. The same low mobility costs make its labor supply more elastic, which is why it pays a larger share of its marginal product.

4.4 Model Fit

First-stage transitions. Table 5 reports the first-stage fit of moments. Mean absolute errors are 0.55 and 0.52 percentage points in the bottom and top markets,

respectively.³⁰ The largest miss is for low human capital workers: the model cannot generate as many of their moves between peripheral firms. This is to be expected since with one switching cost ϕ^{EE} , the log odds of staying rather than switching are the same at every skill level, while in the data they rise with skill, from e.g. 1.97 to 3.01 in the bottom market.

Table 5: First-Stage Fit of Worker Transitions

		Bottom				Top			
		<i>D</i>	<i>P</i>	<i>P'</i>	<i>N</i>	<i>D</i>	<i>P</i>	<i>P'</i>	<i>N</i>
h_1	<i>D</i>	0.39 (0.39)	0.39 (0.39)	0.22 (0.22)		<i>D</i>	0.67 (0.65)	0.18 (0.17)	0.14 (0.17)
	<i>P</i>	0.01 (0.01)	0.67 (0.71)	0.09 (0.05)	0.22 (0.23)	<i>P</i>	0.04 (0.04)	0.68 (0.70)	0.06 (0.03)
	<i>N</i>	0.01 (0.01)	0.44 (0.44)	0.55 (0.55)		<i>N</i>	0.07 (0.07)	0.35 (0.35)	0.58 (0.58)
h_2	<i>D</i>	0.45 (0.45)	0.45 (0.45)	0.10 (0.10)		<i>D</i>	0.74 (0.74)	0.19 (0.19)	0.07 (0.07)
	<i>P</i>	0.01 (0.01)	0.82 (0.82)	0.06 (0.06)	0.11 (0.11)	<i>P</i>	0.05 (0.05)	0.80 (0.80)	0.03 (0.03)
	<i>N</i>	0.01 (0.01)	0.51 (0.51)	0.48 (0.48)		<i>N</i>	0.09 (0.09)	0.43 (0.43)	0.49 (0.49)
h_3	<i>D</i>	0.46 (0.47)	0.42 (0.44)	0.11 (0.10)		<i>D</i>	0.74 (0.75)	0.18 (0.18)	0.08 (0.08)
	<i>P</i>	0.01 (0.01)	0.84 (0.83)	0.04 (0.06)	0.10 (0.10)	<i>P</i>	0.05 (0.05)	0.81 (0.79)	0.03 (0.03)
	<i>N</i>	0.01 (0.01)	0.49 (0.49)	0.50 (0.50)		<i>N</i>	0.09 (0.09)	0.42 (0.42)	0.49 (0.49)

Note: Empirical transition rates from the DADS-Panel, with first-stage model rates in parentheses.

Wages. Table 6 reports the wage fit. Recall that only the dominant firm's wages are targeted in estimation. The mean absolute percentage error for dominant firm wages is 4.05% in the bottom market and 1.20% in the top. Peripheral wages are better matched with the largest miss being 1.49% in the bottom market and 0.23% in the top market.

³⁰ The implied peripheral labor supply elasticities, which are not targeted, range from 2.2 to 2.9 in the bottom market and 2.0 to 2.6 in the top, similar in magnitude to the estimates surveyed in Sokolova and Sorensen (2021).

Table 6: Second-Stage Fit of Wages

	Bottom				Top			
	Peripheral w_P		Dominant w_D		Peripheral w_P		Dominant w_D	
	Data	Model	Data	Model	Data	Model	Data	Model
h_1	23,016	23,195	20,884	20,215	21,259	21,217	26,546	25,983
h_2	35,561	36,092	28,621	30,216	30,478	30,548	35,391	35,771
h_3	38,823	38,622	34,324	33,171	33,519	33,468	39,139	38,974

Note: Wages are in 2015 euros. Model wages are evaluated at χ_{data} .

5 Welfare Costs of Dynamic Monopsony

Having estimated the model, I now quantify the welfare costs of monopsony when labor supply is dynamic. I first report markdowns and discuss the role of future profits in wage setting, then compare the equilibrium allocation with the efficient benchmark. Finally, I ask how a static interpretation of wages changes measured productivity and the welfare cost of monopsony.

5.1 Markdowns

Table 7 reports labor supply elasticities ε , the Lerner ratios $\eta = \varepsilon/(1 + \varepsilon)$, and wages relative to static marginal product ν . Two patterns stand out. First, peripheral firms charge higher markdowns than the dominant firm. Second, at both types of firms and in both markets, the least-skilled workers face the highest markdowns. Peripheral firms behave as static monopsonists, so $\nu_P^h = \eta_P^h$: they pay 70 to 74% of marginal product in the bottom market and 66 to 72% in the top. The dominant firm faces higher labor supply elasticities and pays 88 to 91% and 82 to 85% of static marginal product. Its ratio ν_D^h exceeds η_D^h because hiring also raises future profits: at an interior optimum, Equation (11) gives $\nu_D^h = \eta_D^h[1 + (d^h + c^h)/(p_D h)]$. The gap between the two is therefore not a measure of the welfare loss, which Section 5.2 computes directly.

Table 8 separates the dominant firm's dynamic marginal product of labor into the static marginal product $p_D h$, the continuation term d^h , and the cross-type term c^h defined in Section 2.6. The continuation term is positive at every skill level, adding 2 to 5% of static marginal product in the bottom market and 9 to 11% in the top. Intuitively, raising wages affects the firm's future profits in two ways. First, a worker

Table 7: Labor Supply Elasticities and Markdowns

	Bottom						Top					
	Peripheral			Dominant			Peripheral			Dominant		
	ε_P^h	η_P^h	ν_P^h	ε_D^h	η_D^h	ν_D^h	ε_P^h	η_P^h	ν_P^h	ε_D^h	η_D^h	ν_D^h
h_1	2.30	0.697	0.697	5.64	0.849	0.875	1.95	0.660	0.660	3.02	0.751	0.822
h_2	2.85	0.741	0.741	5.85	0.854	0.894	2.36	0.702	0.702	3.02	0.751	0.835
h_3	2.83	0.739	0.739	8.83	0.898	0.915	2.55	0.719	0.719	3.55	0.780	0.850
Mean	2.65	0.725	0.725	6.65	0.865	0.893	2.29	0.694	0.694	3.19	0.760	0.836

Note: This table reports the labor supply elasticity ε , the Lerner ratio $\eta = \varepsilon/(1 + \varepsilon)$, and the wage relative to static marginal product ν . The markdown equals $1 - \nu$. The mean is weighted by the aggregate distribution of human capital.

hired or retained today produces $p_D h$. If they are retained, they do not pay a switching cost and may have moved up the human capital ladder. This is what I refer to as the *direct effect*, namely the change in future profits from the firm's own workforce, in addition to current production. Second, hiring workers changes the workers employed at rival firms and in non-employment, and with them the future distribution of workers across firms and non-employment. This is what I refer to as the *stock effect*: wage-setting changes the pool from which firms recruit. The dominant firm internalizes these market-wide changes, whereas peripheral firms take the aggregate state as given and do not. Together, these effects yield a positive continuation term d^h at every skill level.³¹ The continuation term is largest for middle-skill workers in both markets. It is larger in the top market, where the dominant firm keeps more of its workers, 67 to 74% each year against 39 to 46% in the bottom (Table 5), and earns a wider margin on each: $p_D h - w_D^h$ is 5,600€ to 7,000€ in the top market and 2,900€ to 3,600€ in the bottom (Tables 6 and 8). The cross-type term is negative but below 100€ in absolute value. The dominant firm therefore pays a larger share of static marginal product than its Lerner ratio η_D^h alone would imply.

³¹ Only their sum is pinned down. Adding a worker to the dominant firm while holding other stocks fixed changes total worker mass.

Table 8: Decomposition of the Dynamic Marginal Product of Labor

	Bottom			Top		
	$p_D h$	d^h	c^h	$p_D h$	d^h	c^h
h_1	23.09	0.74	-0.03	31.61	3.03	-0.06
h_2	33.81	1.60	-0.03	42.82	4.87	-0.09
h_3	36.27	0.70	-0.04	45.83	4.18	-0.05

Note: Units are thousands of 2015 euros.

5.2 Welfare

I now look at the welfare implications of forcing firms to offer the wages that decentralize the efficient allocation, setting $\nu_D^h = \nu_P^h = 1$ for all h . This shuts off all wage-setting power in the market, at both the dominant firm and peripheral firms. I compare the discounted welfare of the decentralized equilibrium path with that of the efficient path, starting both from the distribution observed in the data. Welfare is the planner's objective in Equation (13) and includes production at firms and in non-employment, amenities, taste-shock utility, and switching costs. Table 9 reports the results: welfare is 1.64% lower in the bottom market and 3.47% lower in the top than it would be if all firms paid workers their marginal product. The efficient allocation expands employment and shifts it toward peripheral firms, whose wages initially lie further below marginal product. In the long-run comparison, non-employment falls from 19.2% to 12.9% in the bottom market and from 18.9% to 9.9% in the top. At the same time, the dominant firm's share of employment falls from 2.2% to almost zero in the bottom market and from 22.1% to 2.4% in the top. Paying marginal products therefore raises peripheral wages proportionately more, drawing workers from both non-employment and the dominant firm.

Table 9 also splits the discounted welfare gap into its components. The decentralized economy produces less: firm output is lower by 7.0% of efficient welfare in the bottom market and 9.0% in the top. About half of this is recovered as home production, because the workers monopsony keeps out of employment still produce at home. In the bottom market the planner also pays more switching costs to reallocate workers, which offsets another 1.3 percentage points of the welfare gap. Monopsony is therefore more costly for output than welfare.

Lastly, monopsony reduces human capital accumulation: workers spend less time employed and therefore have fewer opportunities to move up the human capital ladder.

Table 9: Welfare Losses

	Bottom		Top	
Welfare gap (% of planner welfare)	−1.64		−3.47	
Contributions to welfare gap (p.p.)				
Firm output	−7.03		−9.01	
Home production	+3.75		+5.00	
Amenities	+0.29		−0.53	
Taste shocks	+0.10		+1.08	
Switching costs saved	+1.25		−0.01	
Output gap (% of planner output)	−7.67		−9.96	
Annual learning per worker (%)	−8.26		−11.89	
	Decentralized	Planner	Decentralized	Planner
Non-employment (%)	19.19	12.87	18.88	9.87
Dominant share of employment (%)	2.18	0.05	22.12	2.39

Note: Gaps are relative to the planner allocation. Welfare and output are discounted from the observed initial distribution and the five welfare components sum to the total gap. All other outcomes are measured at year 50. Annual learning is new human capital acquired through employment per market worker, before exit and entry. The dominant firm’s employment share excludes non-employed workers.

At year 50, annual learning per worker is 8.3% below its efficient level in the bottom market and 11.9% below in the top. This measures the new human capital acquired through employment in one year, averaged over all workers in the market, before exit and entry. Averaging over all workers captures the learning lost through non-employment.

5.3 The Cost of Ignoring Dynamics

Finally, I examine how interpreting wages through a static wage equation changes inferred productivity and the measured welfare cost of monopsony. I hold the worker-side parameters at their estimates and retain learning, turnover, and the entrant mix.³² I read the dominant firm’s wages as if it were a static monopsonist, setting the continuation term d^h and cross-type term c^h in Equation (11) to zero. Its wage

³² The supply parameters are estimated from worker transitions conditional on observed wages, without using the dominant firm’s wage equation (Section 4.2). Replacing that equation therefore leaves the first-stage estimation problem unchanged.

equation then becomes

$$w_D^h = \frac{\varepsilon_D^h(\chi, w_D)}{\varepsilon_D^h(\chi, w_D) + 1} \times \frac{\partial F_D(\chi, w_D)}{\partial m_D^h(\chi, w_D)} = \eta_D^h p_D h, \quad \forall h.$$

Wages and labor supply elasticities are unchanged, but the inferred productivity is not. At each skill level, the inversion attributes $p_D + (d^h + c^h)/h$ to productivity, overstating productivity by treating the future value of hiring as current output. I invert the equation separately at each skill level using the fitted wages and use the largest implied productivity.³³ I then compute the efficient allocation at this inferred productivity. Keeping worker dynamics unchanged isolates the consequences of the productivity error.³⁴ Table 10 reports the resulting errors in productivity, wage ratios, and welfare losses.

Table 10: The Cost of Ignoring Dynamics

	Bottom		Top	
	Static p_D	Full Model	Static p_D	Full Model
p_D	0.726	0.694	1.094	0.984
$w_D^{h_1}/(p_D h_1)$	0.837	0.875	0.739	0.822
$w_D^{h_2}/(p_D h_2)$	0.854	0.894	0.751	0.835
$w_D^{h_3}/(p_D h_3)$	0.874	0.915	0.765	0.850
Welfare gap (%)	−1.58	−1.64	−2.08	−3.47
Output gap (%)	−7.59	−7.67	−8.95	−9.96
Cost of using static p_D (%)	<0.01	–	0.73	–
Planner non-employment (%)	12.87	12.87	9.86	9.87
Planner dominant share of employment (%)	0.08	0.05	13.10	2.39

Note: Static p_D : productivity inferred from the static wage equation at the fitted wages. Full Model: the estimated model, as in Table 9. Both retain learning, turnover, and the calibrated entrant mix. Welfare and output gaps compare the same decentralized path with the efficient path at each column's productivity. The cost is the welfare lost by implementing the allocation implied by static p_D instead of the full model's efficient allocation. Both allocations are valued at the full model's p_D , and the loss is relative full-model efficient welfare.

³³ The implied values range from 0.71 to 0.73 in the bottom market and from 1.07 to 1.09 in the top. Using each value separately gives an understatement of the welfare loss of 1 to 4% in the bottom market and 35 to 40% in the top.

³⁴ In Appendix B.9, I also remove learning and turnover from the efficient benchmark. The perceived welfare losses fall to 0.50% in the bottom market and 0.72% in the top, understating the full model's losses by 70% and 79%. These larger errors are due to changes in employment, human capital, and the costs of reallocation, in addition to the productivity mismeasurement. Hence to better isolate the costs of interpreting wages through a static wage equation, I keep worker dynamics unchanged in the main exercise.

The static wage equation overstates productivity because a hire can generate profits beyond the current period: an incumbent is cheaper to retain and may accumulate human capital. Given the same wage and labor supply elasticity, the static equation requires higher current productivity to justify that wage. This error is larger in the top market, where the continuation term adds 9 to 11% to static marginal product, against 2 to 5% in the bottom market (Table 8). In the bottom market, the static estimate narrows the dominant firm’s productivity disadvantage but leaves peripheral firms more productive. In the top market, it reverses the ranking: the dominant firm appears more productive than peripheral firms, while the full model’s estimate is slightly below one. Comparing the unchanged wage with this inflated marginal product also makes the markdown appear larger: the inferred markdown is 3.9 to 4.1 percentage points above the full model’s in the bottom market, and 8.3 to 8.5 points above in the top.

The productivity mismeasurement also changes the allocation of workers across firms in the efficient allocation. In the top market, the static estimate assigns 13.1% of employment to the dominant firm, compared with 2.4% at the estimated productivity. Both calculations imply almost the same non-employment rate, so the difference is in where workers should work. The static estimate calls for much less reallocation toward peripheral firms. Evaluated at the full model’s productivity, implementing that allocation would forgo 21% of the available welfare gain, or 0.73% of full-model efficient welfare. In the bottom market, the dominant firm accounts for just 2.2% of employment, compared with 22% in the top. The smaller productivity error leaves its productivity well below that of peripheral firms, so both estimates imply nearly the same efficient allocation.

By making the dominant firm’s large workforce appear more efficient, the static estimate reduces the perceived welfare gain from reallocation. I value both the decentralized path in Table 9 and the efficient allocation at the inferred productivity. The perceived welfare losses are 1.58% and 2.08%, against 1.64% and 3.47% in the full model: the static wage equation understates the loss by 4% in the bottom market and 40% in the top, or 22% on average across the two markets. The two measurement errors therefore go in opposite directions. The same productivity error makes wages appear further below marginal product and the allocation appear closer to efficient. Here, ignoring dynamics leads to a larger measured markdown but a smaller estimated welfare loss.

6 Conclusion

In this paper, I study how the welfare costs of monopsony change when accounting for endogenous human capital accumulation. In doing so, I make several contributions to the existing literature. Motivated by novel descriptive evidence on local labor markets in France and their competitive structure, I develop a general equilibrium model of monopsonistic labor markets. The model features worker heterogeneity in the form of general human capital and a finite number of firms that compete for workers. Importantly, the model has a non-atomistic firm that internalizes the effect its wage-setting has on the aggregate distribution of human capital and the future labor supply it faces. I then study a planner who maximizes market-wide welfare by allocating workers across firms and non-employment and characterize the wages that decentralize the efficient outcome. Because the dominant firm’s wage equation contains a dynamic term, its labor supply elasticity no longer pins down its markdown, and welfare has to be assessed with the equilibrium allocations and the planner’s problem. I solve the recursive equilibrium globally using deep learning. In the estimated markets, monopsony reduces employment and human capital accumulation and allocates too many workers to the dominant firm. Inferring productivity from the static wage equation overstates the dominant firm’s productivity, making its markdown appear larger while understating the welfare cost of monopsony.

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A Data Appendix

A.1 Data Sources and Sample Construction

I use two French administrative matched employer-employee datasets from the *Déclarations annuelles de données sociales* (DADS):

Short panel. The first dataset I use is the DADS-Postes, which covers the near-universe of jobs in France. This is a spell-based dataset, meaning there is one observation per individual-job-year.³⁵ Hence a worker with two jobs appears as two distinct observations. However, the worker IDs are reshuffled every two years, making it impossible to match workers across time for more than two periods. I use this to document descriptive evidence on French labor markets which I present in Section 1.2 and Section 1.3.

Long panel. The second dataset I use is the DADS-Panel. It contains the full employment history of all workers born in October of even years before 2002 and of every year from 2003 onwards. I treat it as a representative sample of the French labor force which can be used for longitudinal analysis.³⁶ Once a worker enters the dataset, all her subsequent employment spells are observable. I use this to measure human capital when I estimate the model.

Both sources are at the annual frequency and contain roughly the same worker-level variables. These include detailed occupation codes, industry codes, zipcodes of residence and work, wages, hours, and demographics such as age and gender. However, the DADS has several drawbacks. The main disadvantage in the context of this study is that worker education is not observable. Instead, I use other worker characteristics to estimate unobserved human capital, which I describe in detail in Section 4.1. Another limitation is the absence of firm balance sheet data. While many firm and establishment level information such as wage bills and employment are available, data on revenue or input choices is not.³⁷ Finally, since workers only

³⁵ For example, if a worker has two jobs in a given year, they appear as two separate observations with the same worker ID.

³⁶ There are birth month effects in labor markets as well as seasonality patterns to births that could in principle induce non-random selection into this subsample.

³⁷ Firm balance sheet information for French firms exists (the FARE/FICUS datasets), but due to

appear in the data when they are employed, we cannot distinguish between non-employment and unemployment. For this reason, I do not model unemployment and restrict workers to be either employed or non-employed.

Sample construction. I apply the same sample construction rules for both datasets. I focus on workers aged 16-65 who are employed in the private sector in metropolitan France. For the DADS-Postes, I use the repeated cross-section of 2014-2015 (henceforth DADS-Postes 2015). For the DADS-Panel, I go back to 2003,³⁸ the first year following a major change in occupation codes, up until 2016. An additional period after 2015 is needed to identify non-employed workers in 2015.

A.2 Market Definition

This appendix collects the statistics behind the market definition in Section 1.2: self-flow rates across segmentations, summary statistics for occupations and commuting zones, and an example of the granularity of the 4-digit occupation codes.

Table A.1: Self-Flow Rates

	Rate (%)
Region	97.06
Department	93.15
Commuting Zone (CZ)	92.62
4-digit occ	74.69
4-digit occ $\times$ CZ	72.25

Note: Calculated in the DADS-Postes from 2014 to 2015.

legal restrictions is not available to researchers in North America.

³⁸ The DADS-Panel is also bigger after 2003. Before 2003, only workers born in October of even years are in the dataset, whereas after (and including) 2003, every worker born in October is included.

Table A.2: Occupation and Commuting Zone Summary Statistics

	Mean	SD	p25	Median	p75	p95
Occupations						
Workers/ 2-digit occ	567,448	573,765	32,692	333,143	1,122,509	1,751,103
Workers/ 4-digit occ	43,520	62,853	8,824	23,462	46,395	177,993
2-digit occ/firm (unweighted)	2.18	1.93	1	1	3	6
2-digit occ/firm (weighted)	3.66	6.58	1	1	1	22
4-digit occ/firm (unweighted)	2.47	4.64	1	1	2	8
4-digit occ/firm (weighted)	183.1	90.8	103	160	266	304
Number of 2-Digit Occupations	27					
Number of 4-Digit Occupations	384					
Commuting Zones						
Workers/CZ	54,972	163,860	12,038	21,783	51,545	154,896
Firms/CZ (weighted)	211,625	100,488	267,793	267,793	267,793	267,793
4-digit occ/CZ (weighted)	383	5	384	384	384	384
2-digit occ/CZ (weighted)	27	0.4	27	27	27	27
Number of CZs	304					

Note: Occupation rows weighted by firm employment; commuting zone rows weighted by commuting zone employment. Computed in the DADS-Postes 2015.

Table A.3: Granularity of PCS-ESE occupation codes

4: Intermediate Professions. The 2-digit occupations are 42 Schoolteachers, instructors and similar; 43 Intermediate health and social work professions; 44 Clergy, religious; 45 Intermediate administrative professions of the public service; 46 Intermediate administrative and commercial professions of companies; 47 Technicians (except service technicians); 48 Foremen and supervisors.

4-digit occupations within code 43	
431a Nursing executives	434a Socio-educational intervention managers
431b Psychiatric nurses	434b Social work assistants
431c Childcare workers	434c Family counselors
431d Specialized nurses (other than psychiatric and pediatric)	434d Specialized educators
431e Midwives (salaried)	434e Support workers
431f Nurses in general care (salaried)	434f Specialized technical educators, workshop instructors
432b Rehabilitation massage therapists (salaried)	434g Early childhood educators
432d Other rehabilitation specialists (salaried)	435a Directors of socio-cultural and leisure centers
433a Medical technicians	435b Socio-cultural and leisure organizers
433b Eyewear opticians and audioprosthodontists (salaried)	
433c Other medical device specialists (salaried)	
433d Pharmacy assistants	

Note: PCS-ESE classification from DADS. Full classification: <https://www.insee.fr/fr/information/2497952>.

A.3 Leader Statistics

This appendix reports summary statistics for the leader’s size relative to other firms.

Table A.4: Leader Summary Statistics

	Mean	SD	p25	Median	p75	p95
Leader Share (unweighted)	31	21	14	26	45	75
Leader Share (weighted)	16	18	4	10	22	55
Leader Share / Second Share (unweighted)	2.9	6.5	1.3	1.8	2.5	7.3
Leader Share / Second Share (weighted)	2.8	8.3	1.2	1.5	2.3	6.8
Mean Non-Leader Share (unweighted)	11.1	11.1	3.7	7.5	14.3	33.3
Mean Non-Leader Share (weighted)	3.3	4.6	0.6	1.7	4.0	12.0
Leader Share / Mean Non-Leader Share (unweighted)	5.0	9.3	2.0	3.0	5.0	13.3
Leader Share / Mean Non-Leader Share (weighted)	10.1	18.2	3.7	5.8	10.1	28.0

Note: Unweighted statistics computed with equal weights for each market. Weighted statistics use market employment as weights. Markets with only one firm (leader’s employment share = 100%) are excluded.

A.4 Leader-Share Persistence

In this section, I analyze the persistence of leader shares over time. For every market i and year t from 2003 to 2016, I compute the leader’s employment share $L_{i,t}$ in the DADS-Panel and estimate

$$L_{i,t} = \beta_0 + \beta_1 t + \beta_2 \text{Low}_i + \beta_3 (t \times \text{Low}_i) + \epsilon_{i,t} \quad (\text{A.1})$$

where Low_i indicates a below-median leader share. This dataset samples 1/12th of workers, so the median number of sampled workers per market-year $n_{i,t}$ is 4 and grows by 2.3% per year. Since $L_{i,t} \geq 1/n_{i,t}$, $L_{i,t}$ trends down even if the true share is constant, and in small cells the sampled leader need not be the true leader. To address this, I also restrict the sample to cells with at least 20 sampled workers and control for $1/n_{i,t}$ in some specifications.

Table A.5 reports the estimates. In the full panel, the leader’s share declines by 0.69 percentage points per year in concentrated markets and by 1.38 in unconcentrated markets. Restricting the sample to cells with at least 20 sampled workers cuts the latter to 0.45 and flips the gradient. With market fixed effects and a control for $1/n_{i,t}$, the annual declines are 0.39 and 0.63 percentage points in concentrated

and unconcentrated markets, respectively. I therefore treat the leader's position as permanent and estimate the model at the observed distribution of workers across firms.

Table A.5: Leader-Share Trend and the Sampling Floor

	Year		Year $\times$ Low		N
Full panel	-0.0069	(0.0001)	-0.0069	(0.0002)	1,105,848
Cells ≥ 20 workers	-0.0082	(0.0003)	+0.0037	(0.0003)	146,786
Market FE + $1/n_{i,t}$	-0.0039	(0.0001)	-0.0024	(0.0001)	1,105,848

Note: Dependent variable is leader employment share by market-year from DADS-Panel 2003–2016. Low is below-median leader share. Standard errors are clustered by market.

B Model Appendix

B.1 Worker Values and Choice Probabilities

$u(h, o, \chi, d)$ is the contemporaneous utility received if alternative d is chosen. This is the sum of wages and the amenity value of working minus the relevant switching costs:

$$\begin{aligned} u(h, o, \chi, D) &= w_D^h(\chi) + a_D^h - \phi^{NE} \mathbb{1}\{o = N\} - \phi^{PD} \mathbb{1}\{o = P\}, \\ u(h, o, \chi, P) &= w_P^h(\chi) - \phi^{NE} \mathbb{1}\{o = N\} - \phi^{DP} \mathbb{1}\{o = D\} - \phi^{EE} \mathbb{1}\{o = P'\}, \end{aligned}$$

where $P' \neq P$ is a different peripheral firm. Peripheral firms post the same wages and differ to a worker only through ϕ^{EE} . The choice-specific value for non-employment $d = N$ is

$$v(h, o, \chi, N) = u(h, o, \chi, N) + \beta(1 - \mu) \mathbb{E}_{h'|h, N} V(h', N, \chi')$$

where $u(h, o, \chi, N) = b^h$, $\forall o$ since a non-employed worker only derives utility from their home production. Under the assumption that ϵ is Type 1 Extreme Value, the ex-ante value of a worker with origin o has the closed form

$$V(h, o, \chi) = \sigma \left(\ln \sum_{j \in \{D, y_1, \dots, y_M, N\}} \exp \frac{v(h, o, \chi, j)}{\sigma} + \gamma^{EM} \right), \quad (\text{B.1})$$

where $\gamma^{EM} \approx 0.5772$ is the Euler-Mascheroni constant and the sum runs over individual firms. For $o \in \{D, N\}$, the M peripheral terms are identical; for $o = P$, one term is the worker's current firm and the other $M - 1$ equal $v(h, P, \chi, P) - \phi^{EE}$.

The set of $10 \times H$ choice probabilities are given by

$$P(d|h, o, \chi) = \int \mathbb{1}\{d = \arg \max_{j \in \{D, y_1, \dots, y_M, N\}} [v(h, o, \chi, j) + \epsilon_j]\} g(d\epsilon|h, o, \chi).$$

where $\mathbb{1}$ is the indicator function. The Type 1 Extreme Value assumption yields the logit probabilities in Equation (5).

B.2 Worker Flows

I omit time subscripts for ease of exposition. At the beginning of the period, the measures of workers with human capital h at the dominant firm, at the peripheral firm y , and in non-employment are $m_{D,-1}^h$, $m_{P,-1}^{h,y}$, and $m_{N,-1}^h$, respectively, where the -1 subscript indicates that this distribution is inherited from the previous period. First, worker human capital evolves according to Equation (2). The resulting measures of workers with human capital h are given by

$$\hat{e}_D^h = m_{D,-1}^h (1 - g_E \mathbb{1}_{\{h < h_H\}}) + m_{D,-1}^{h-1} g_E \mathbb{1}_{\{h > h_1\}}, \quad (\text{B.2})$$

$$\hat{e}_P^{h,y} = m_{P,-1}^{h,y} (1 - g_E \mathbb{1}_{\{h < h_H\}}) + m_{P,-1}^{h-1,y} g_E \mathbb{1}_{\{h > h_1\}}, \quad \forall y \in \mathcal{P} \quad (\text{B.3})$$

$$\hat{n}^h = m_{N,-1}^h \quad (\text{B.4})$$

where $\mathbb{1}$ is the indicator function. Next, exogenous market turnover removes a fraction μ of workers and an equal mass μ of entrants replaces them. I assume the new workers enter the market employed: a share ξ^h of them has human capital h , and among level- h entrants, a share ξ_D^h joins the dominant firm while the rest spread evenly across the peripheral firms. The post-turnover measures are

$$\begin{aligned} \tilde{e}_D^h &= (1 - \mu) \hat{e}_D^h + \mu \xi^h \xi_D^h, & \tilde{n}^h &= (1 - \mu) \hat{n}^h, & \forall h \in \mathcal{H}, \\ \tilde{e}_P^{h,y} &= (1 - \mu) \hat{e}_P^{h,y} + \mu \xi^h \frac{1 - \xi_D^h}{M}, & & \forall h \in \mathcal{H}, \forall y \in \mathcal{P}. \end{aligned} \quad (\text{B.5})$$

Finally, employed workers are exogenously separated into non-employment at rate δ^h . A worker separated this period does *not* participate in the current period's choice

stage: the separation places them in non-employment, where they spend the period earning b^h and only choose a destination at $t + 1$. The post-separation measures are

$$e_D^h = (1 - \delta^h) \tilde{e}_D^h, \quad \forall h \in \mathcal{H} \quad (\text{B.6})$$

$$e_P^{h,y} = (1 - \delta^h) \tilde{e}_P^{h,y}, \quad \forall h \in \mathcal{H}, \forall y \in \mathcal{P} \quad (\text{B.7})$$

$$n^h = \tilde{n}^h + \delta^h \tilde{e}_D^h + \delta^h \sum_{y \in \mathcal{P}} \tilde{e}_P^{h,y}, \quad \forall h \in \mathcal{H}. \quad (\text{B.8})$$

so the non-employment stock n^h splits into the $\tilde{n}^h$ who choose where to go this period and the just-separated $\delta^h \tilde{e}_D^h + \delta^h \sum_{y \in \mathcal{P}} \tilde{e}_P^{h,y}$, who cannot choose until $t + 1$. The firms then choose wages to maximize the present discounted value of profits. I denote the aggregate distribution of workers across firms and non-employment at the time of their choice as $(e_P^{y_1}, \dots, e_P^{y_M}, e_D, n) \subset \Delta^{(M+2) \times H}$ where $e_P^y := (e_P^{1,y}, e_P^{2,y}, \dots, e_P^{H,y})$, $\forall y \in \mathcal{P}$ is the workforce of peripheral firm y , $e_D := (e_D^1, e_D^2, \dots, e_D^H)$ is the workforce of the dominant firm, and $n := (n^1, n^2, \dots, n^H)$ is the distribution of non-employed workers. The firms' states are slices of this aggregate distribution, which I define in Section 2.3. Workers then observe their taste shocks ϵ and decide which firm to work for or non-employment. Firms set wages before the shocks are realized, so they anticipate the choice probabilities. This mobility is characterized by conditional choice probabilities, which yield the distribution at the end of the period m_D, m_P^y , and m_N . Finally, after matches have been consummated, production occurs both at home and on the job. Then the next period begins.

B.3 The Planner's Problem

Consider a social planner that chooses the allocation of workers across firms to maximize market-wide welfare. Consistent with the decentralized equilibrium, I assume the planner's problem occurs at the same time as firms choose wages, i.e. after human capital changes and before workers receive their taste shocks (see Figure 3 for model timing). Because of the switching costs, the aggregate distribution of human capital is not sufficient and the planner needs to know the distribution of workers across the dominant firm, the peripheral firms, and non-employment. His state is thus also $\chi = (e_D, n, s) \subset \Delta^{3 \times H}$. The planner simultaneously chooses the allocation of all worker types to a destination d . In the model, workers go where they receive the most utility. Here, the planner chooses a value to offer each worker type (h, o, ϵ)

to affect their mobility choice. Notice that since the utility value workers receive from their idiosyncratic taste shock ϵ affects their choice of destination, the planner's choice of value is indexed by ϵ . As in [Lagakos, Mobarak, and Waugh \(2023\)](#) and [Waugh \(2023\)](#), this is equivalent to the planner directly choosing the $10 \times H$ probabilities $\{P_{D \rightarrow d}^h\}_{d \in \{D, P, N\}}, \{P_{P \rightarrow d}^h\}_{d \in \{D, P, P', N\}}, \{P_{N \rightarrow d}^h\}_{d \in \{D, P, N\}}, \forall h \in \mathcal{H}$, which I collapse into $\mathbb{P} \subset [0, 1]^{10 \times H}$. Note these are probabilities so there are the same three associated constraints that ensure they sum to 1 as in the decentralized equilibrium.

The planner's choice of $\mathbb{P}$ determines worker mobility within the period. The number of workers with human capital h the planner sends to the dominant firm is

$$m_{D,S}^h(\mathbb{P}) := e_D^h P_{D \rightarrow D}^h + s^h P_{P \rightarrow D}^h + \tilde{n}^h P_{N \rightarrow D}^h. \quad (\text{B.9})$$

Similarly, the number of workers with human capital h in non-employment is

$$m_{N,S}^h(\mathbb{P}) := e_D^h P_{D \rightarrow N}^h + s^h P_{P \rightarrow N}^h + \tilde{n}^h P_{N \rightarrow N}^h + \delta^h \tilde{e}_D^h + \delta^h \sum_{y \in \mathcal{P}} \tilde{e}_P^{h,y}. \quad (\text{B.10})$$

As in the decentralized economy, the non-employment stock splits into the choosers $\tilde{n}^h$, whom the planner can allocate this period, and the just-separated waiters $\delta^h \tilde{e}_D^h + \delta^h \sum_{y \in \mathcal{P}} \tilde{e}_P^{h,y}$, who are forced into non-employment for one period and so enter welfare only through their home production b^h (they draw no taste shock, pay no switching cost, and earn no amenity). The planner faces the same constraint as the decentralized economy. Next, the inflow of workers at any of the M peripheral firms includes workers coming from the dominant firm and non-employment, workers remaining at the peripheral firm, and workers coming from the other $M - 1$ peripheral firms is

$$m_{P,S}^h(\mathbb{P}) := e_D^h P_{D \rightarrow P}^h + \frac{s^h}{M} P_{P \rightarrow P}^h + \frac{M-1}{M} s^h P_{P' \rightarrow P}^h + \tilde{n}^h P_{N \rightarrow P}^h. \quad (\text{B.11})$$

Since the planner allocates all workers somewhere, we have the adding-up constraint $\sum_h m_{D,S}^h(\mathbb{P}) + m_{N,S}^h(\mathbb{P}) + M m_{P,S}^h(\mathbb{P}) = 1$. Finally, because switching costs differ by transition, it is convenient to define the masses of workers with human capital h making each type of firm-to-firm move,

$$\begin{aligned} m_{D \rightarrow P,S}^h(\mathbb{P}) &:= M e_D^h P_{D \rightarrow P}^h, & m_{P \rightarrow D,S}^h(\mathbb{P}) &:= s^h P_{P \rightarrow D}^h, \\ m_{P \rightarrow P',S}^h(\mathbb{P}) &:= s^h (M-1) P_{P \rightarrow P'}^h, \end{aligned} \quad (\text{B.12})$$

as well as the number of workers who transition out of non-employment,

$$m_{N \rightarrow E, S}^h(\mathbb{P}) := \tilde{n}^h P_{N \rightarrow D}^h + M \tilde{n}^h P_{N \rightarrow P}^h. \quad (\text{B.13})$$

To simplify notation, as before, I collapse these into the H -dimensional vectors $m_{D, S}, m_{N, S}, m_{P, S}, m_{D \rightarrow P, S}, m_{P \rightarrow D, S}, m_{P \rightarrow P', S}$, and $m_{N \rightarrow E, S}$.

The utility workers derive from the taste shocks is determined by the planner's choice. Hence, a key component of the planner's objective is the *ex-ante* (expected) value of the taste shock for all workers. The Type 1 Extreme Value assumption yields a closed-form expression for this object, commonly referred to as the conditional choice expected utility (see [Aguirregabiria 2021](#) section 11.2 for a derivation). Given a transition probability $P_{o \rightarrow d}^h$ chosen by the planner, the expected value of the worker's preference shock is $\mathbb{E}[\epsilon | P_{o \rightarrow d}^h] = \sigma \gamma^{EM} - \sigma \ln P_{o \rightarrow d}^h$, where, again, γ^{EM} is the Euler-Mascheroni constant. Aggregating over all workers, the total ex-ante value of these shocks given the planner's choice of $\mathbb{P}$ is

$$\begin{aligned} \mathcal{E}(\mathbb{P}) := & \sum_h e_D^h \left(P_{D \rightarrow D}^h \mathbb{E}[\epsilon | P_{D \rightarrow D}^h] + M P_{D \rightarrow P}^h \mathbb{E}[\epsilon | P_{D \rightarrow P}^h] + P_{D \rightarrow N}^h \mathbb{E}[\epsilon | P_{D \rightarrow N}^h] \right) \\ & + \sum_h s^h \left(P_{P \rightarrow D}^h \mathbb{E}[\epsilon | P_{P \rightarrow D}^h] + P_{P \rightarrow P}^h \mathbb{E}[\epsilon | P_{P \rightarrow P}^h] + (M-1) P_{P \rightarrow P'}^h \mathbb{E}[\epsilon | P_{P \rightarrow P'}^h] + P_{P \rightarrow N}^h \mathbb{E}[\epsilon | P_{P \rightarrow N}^h] \right) \\ & + \sum_h \tilde{n}^h \left(P_{N \rightarrow D}^h \mathbb{E}[\epsilon | P_{N \rightarrow D}^h] + M P_{N \rightarrow P}^h \mathbb{E}[\epsilon | P_{N \rightarrow P}^h] + P_{N \rightarrow N}^h \mathbb{E}[\epsilon | P_{N \rightarrow N}^h] \right). \end{aligned}$$

The first term is the total value of the shocks for workers that begin the period employed at the dominant firm. There are e_D^h of these workers with human capital h . Of those, a fraction $P_{D \rightarrow D}^h$ will remain at the dominant firm, a fraction $M P_{D \rightarrow P}^h$ will move to a peripheral firm, and a fraction $P_{D \rightarrow N}^h$ will move to non-employment. Within each of these groups, workers will receive different vectors of taste shocks, but their expected taste shock will be the same and given by the expression above. The second term and last term are analogous expressions for workers starting the period employed at peripheral firms and non-employed, respectively.

The planner's objective is given by Equation (13), subject to these worker flows, the law of motion in Appendix B.2, and the three adding-up constraints for $\mathbb{P}$.

B.4 Proof of the Decentralization Theorem

Proof of Theorem 1. Production is linear, and a worker's amenities, switching costs, and human capital and employment transitions do not depend on other workers. The planner's problem is therefore separable across workers and the planner can choose each worker's destination to maximize the worker's expected discounted contribution to welfare.³⁹ Let $S(h, o, \chi)$ denote this value before taste shocks are realized for a worker who can choose a destination this period.⁴⁰ As in Equation (3), it satisfies

$$S(h, o, \chi) = \int \max_d \left[s(h, o, \chi, d) + \epsilon_d \right] g(d\epsilon|h, o, \chi) \quad (\text{B.14})$$

where $s(h, o, \chi, d)$ is the choice-specific welfare (excluding taste shock ϵ_d). As in the decentralized equilibrium, workers allocated to employment destinations face the exogenous separation shock $\delta^{h'}$ after human capital evolves. Define $U_S(h, \chi)$ as the welfare when separated into non-employment:

$$U_S(h, \chi) = b^h + \beta(1 - \mu)\mathbb{E}_{h'|h, N} S(h', N, \chi'). \quad (\text{B.15})$$

The choice-specific welfare values are then given by

$$\begin{aligned} s(h, o, \chi, d) &= u_S(h, o, \chi, d) + \beta(1 - \mu)\mathbb{E}_{h'|h, d} \left[\delta^{h'} U_S(h', \chi') + (1 - \delta^{h'}) S(h', d, \chi') \right], \quad \forall d \in \{D, P\} \\ s(h, o, \chi, N) &= u_S(h, o, \chi, N) + \beta(1 - \mu)\mathbb{E}_{h'|h, N} S(h', N, \chi') \end{aligned}$$

where $u_S(h, o, \chi, d)$ is the contemporaneous welfare (excluding the realization of the taste shock ϵ_d) resulting from a worker of human capital h with origin o going to a destination d :

$$\begin{aligned} u_S(h, o, \chi, D) &= \frac{\partial F_D}{\partial m_D^h} + a_D^h - \phi^{NE} \mathbb{1}\{o = N\} - \phi^{PD} \mathbb{1}\{o = P\}, \\ u_S(h, o, \chi, P) &= \frac{\partial F_P}{\partial m_P^h} - \phi^{NE} \mathbb{1}\{o = N\} - \phi^{DP} \mathbb{1}\{o = D\} - \phi^{EE} \mathbb{1}\{o = P\}, \\ u_S(h, o, \chi, N) &= b^h. \end{aligned}$$

³⁹ Workers waiting after a separation are not allocated this period, and future entrants arrive exogenously. Neither depends on today's allocation, so both add a constant to welfare.

⁴⁰ To be precise, this value does not depend on χ , but I write it this way to facilitate comparison with the worker's problem. The planner value $S(\chi)$ still depends on the distribution of workers across states.

With linear production, the marginal products are $\frac{\partial F_D}{\partial m_D^h} = p_D \times h$ and $\frac{\partial F_P}{\partial m_P^h} = h$ since p_P is normalized to 1. If wages equal these marginal products in every period, each worker's current payoff equals their contribution to current welfare. Their skill transitions, separation risks, switching costs, and survival-adjusted discount factor $\beta(1 - \mu)$ are also the same so the two problems consequently coincide. Thus the wage schedules $w_{D,S}^h = p_D h$ and $w_{P,S}^h = h$ implement the efficient allocation. $\square$

B.5 Dominant Firm FOCs

The dominant firm chooses the full vector $w_D = (w_D^1, \dots, w_D^H)$ to maximize profits. Differentiating Equation (10) with respect to w_D^h and using $\partial F_D / \partial m_D^{h'} = p_D h'$ yields the following first-order condition for each h :

$$\sum_{h'} p_D h' \frac{\partial m_D^{h'}(\chi, w_D)}{\partial w_D^h} - m_D^h(\chi, w_D) - \sum_{h'} \frac{\partial m_D^{h'}(\chi, w_D)}{\partial w_D^h} w_D^{h'} + \beta \frac{\partial \mathbb{E} J_D(\chi')}{\partial w_D^h} = 0 \quad (\text{B.16})$$

A change in any one wage moves output and the wage bill through the labor supply responses of *all* worker types, and it moves the continuation value through next period's distribution. To reduce clutter, I suppress the arguments (χ, w_D) in what follows. The choice of wages affects χ' only through the labor supply curves entering the laws of motion, so the chain rule routes the continuation derivative through the same supply responses:

$$\beta \frac{\partial \mathbb{E} J_D(\chi')}{\partial w_D^h} = \sum_{h'} \frac{\partial m_D^{h'}}{\partial w_D^h} d^{h'}, \quad (\text{B.17})$$

Substituting Equation (B.17) into Equation (B.16), the three sums share the common factor $\partial m_D^{h'} / \partial w_D^h$ and combine to

$$\sum_{h'} \frac{\partial m_D^{h'}}{\partial w_D^h} \left(p_D h' - w_D^{h'} + d^{h'} \right) = m_D^h. \quad (\text{B.18})$$

Separating the own-type term from the cross-type terms,

$$\frac{\partial m_D^h}{\partial w_D^h} \left(p_D h - w_D^h + d^h \right) + \sum_{h' \neq h} \frac{\partial m_D^{h'}}{\partial w_D^h} \left(p_D h' - w_D^{h'} + d^{h'} \right) = m_D^h,$$

and dividing through by $\partial m_D^h / \partial w_D^h > 0$ gives

$$p_D h - w_D^h + d^h + c^h = \frac{m_D^h}{\partial m_D^h / \partial w_D^h}$$

where I refer to $c^h := \sum_{h' \neq h} \frac{\partial m_D^{h'} / \partial w_D^h}{\partial m_D^h / \partial w_D^h} (p_D h' - w_D^{h'} + d^{h'})$ as the cross-type term. This is the surplus on the workers of other types that the firm attracts or repels, per type- h worker attracted directly. Equation (6) implies $m_D^h / (\partial m_D^h / \partial w_D^h) = w_D^h / \varepsilon_D^h$, so the previous equation becomes

$$p_D h - w_D^h + d^h + c^h = \frac{w_D^h}{\varepsilon_D^h} \iff w_D^h \frac{\varepsilon_D^h + 1}{\varepsilon_D^h} = p_D h + d^h + c^h,$$

and solving for w_D^h yields Equation (11) in the main text, where $p_D h$ is the static component of the marginal product.

The off-diagonal supply responses, $\partial m_D^{h'} / \partial w_D^h$ for $h' \neq h$, and with them the cross-type term, arise exclusively through the non-atomistic channel: a deviation in w_D^h moves χ' , which moves the continuation value of every worker type, and hence every type's choice probabilities (Appendix B.6). In the static model ($\beta = 0$) they vanish identically, since the choice-specific values of a type- h' worker contain only $w_D^{h'}$, and Equation (B.18) reduces to the textbook monopsony first-order condition.

B.6 Mechanics of non-atomistic dominant firm

In this section, I illustrate what it means for the dominant firm to be non-atomistic. In choosing wages, the dominant firm fully internalizes its effect on the sorting of workers, not only to itself but also to rival peripheral firms and non-employment. This works through the choice probabilities and shows up in two related ways in the firm's first-order conditions presented in Appendix B.5. First, the number of workers that decide to come changes as the firm varies its wages. Differentiating Equation (7), this change is given by

$$\frac{\partial m_D^h(\chi, w_D)}{\partial w_D^h} = e_D^h \frac{\partial P_{D \rightarrow D}^h}{\partial w_D^h} + s^h \frac{\partial P_{P \rightarrow D}^h}{\partial w_D^h} + \tilde{n}^h \frac{\partial P_{N \rightarrow D}^h}{\partial w_D^h}.$$

Second, the policy choice today affects the continuation value through the distribution the firm will face tomorrow. This effect can be computed using the chain rule:

$$\frac{\partial J_D(\chi')}{\partial w_D^h} = \sum_{\hat{h}} \left[\frac{\partial J_D(\chi')}{\partial (e_D^{\hat{h}})'} \frac{\partial (e_D^{\hat{h}})'}{\partial w_D^h} + \frac{\partial J_D(\chi')}{\partial (s^{\hat{h}})'} \frac{\partial (s^{\hat{h}})'}{\partial w_D^h} + \frac{\partial J_D(\chi')}{\partial (n^{\hat{h}})'} \frac{\partial (n^{\hat{h}})'}{\partial w_D^h} \right].$$

The first terms can all be computed using the envelope theorem. The dominant firm's value next period is

$$J_D(\chi') = \max_{w_D' \in \mathbb{R}_+^H} p_D \sum_{h \in \mathcal{H}} (m_D^h)' h - \sum_h (m_D^h)' (w_D^h)' + \beta \mathbb{E} J_D(\chi'')$$

where

$$(m_D^h)' = (e_D^h)' (P_{D \rightarrow D}^h)' + (s^h)' (P_{P \rightarrow D}^h)' + (\tilde{n}^h)' (P_{N \rightarrow D}^h)'$$

and χ_D'' evolves according to the laws of motion in Appendix B.2. So, for example, the change in the value next period given a change in the number of workers with human capital h at the dominant firm next period is

$$\begin{aligned} \frac{\partial J_D(\chi')}{\partial (e_D^h)'} &= \left[(P_{D \rightarrow D}^h)' - \frac{\delta^h}{1 - \delta^h} (P_{N \rightarrow D}^h)' \right] (p_D h - (w_D^h)') \\ &\quad + \sum_{\hat{h}} (p_D \hat{h} - (w_D^{\hat{h}})') \left[(e_D^{\hat{h}})' \frac{\partial (P_{D \rightarrow D}^{\hat{h}})'}{\partial (e_D^{\hat{h}})'} + (s^{\hat{h}})' \frac{\partial (P_{P \rightarrow D}^{\hat{h}})'}{\partial (e_D^{\hat{h}})'} + (\tilde{n}^{\hat{h}})' \frac{\partial (P_{N \rightarrow D}^{\hat{h}})'}{\partial (e_D^{\hat{h}})'} \right] \\ &\quad + \beta \mathbb{E} \left[\frac{\partial J_D(\chi'')}{\partial \chi''} \frac{\partial \chi''}{\partial (e_D^h)'} \right]. \end{aligned}$$

The first term is the direct atomistic effect: the firm earns a positive margin $p_D h - (w_D^h)'$ on each retained worker. It is net of the waiters: with n^h fixed, an extra worker at the dominant firm adds $\delta^h / (1 - \delta^h)$ separated workers who wait this period, and so removes that many choosers from $\tilde{n}^h = n^h - \frac{\delta^h}{1 - \delta^h} (e_D^h + s^h)$. The second term is the *stock effect* I discuss in the main text: an extra type- h worker shifts the whole distribution χ' . The stock terms can be negative, so the overall sign of $\partial J_D(\chi') / \partial (e_D^h)'$ is ambiguous and ultimately a quantitative question. The rest of the terms can also be computed using the laws of motion. Again, this all works through the conditional choice probabilities.

I now show how firms affect worker mobility through the conditional choice probabilities. The choice of wages affects all choice probabilities. For ease of exposition, I

only report how $P_{D \rightarrow D}^h$ changes when the dominant firm raises wages. This can easily be computed as

$$\begin{aligned} \frac{\partial P_{D \rightarrow D}^h}{\partial w_D^h} &= \frac{e^{v(h,D,\chi,D)/\sigma} \frac{\partial v(h,D,\chi,D)}{\partial w_D^h}}{\sigma \left(e^{v(h,D,\chi,D)/\sigma} + M e^{v(h,D,\chi,P)/\sigma} + e^{v(h,D,\chi,N)/\sigma} \right)} \\ &\quad - \frac{e^{v(h,D,\chi,D)/\sigma} \left(e^{v(h,D,\chi,D)/\sigma} \frac{\partial v(h,D,\chi,D)}{\partial w_D^h} + M e^{v(h,D,\chi,P)/\sigma} \frac{\partial v(h,D,\chi,P)}{\partial w_D^h} + e^{v(h,D,\chi,N)/\sigma} \frac{\partial v(h,D,\chi,N)}{\partial w_D^h} \right)}{\sigma \left(e^{v(h,D,\chi,D)/\sigma} + M e^{v(h,D,\chi,P)/\sigma} + e^{v(h,D,\chi,N)/\sigma} \right)^2} \\ &= \frac{1}{\sigma} P_{D \rightarrow D}^h \left[(1 - P_{D \rightarrow D}^h) \frac{\partial v(h,D,\chi,D)}{\partial w_D^h} - M P_{D \rightarrow P}^h \frac{\partial v(h,D,\chi,P)}{\partial w_D^h} - P_{D \rightarrow N}^h \frac{\partial v(h,D,\chi,N)}{\partial w_D^h} \right] \end{aligned}$$

which takes on the usual form (e.g. [Train 2009](#) section 3.6). However, there is one key feature here that separates this from standard dynamic discrete choice models: the supply (usually demand in IO models) is dynamic. Indeed, recall that the choice-specific worker value in this case is

$$v(h, D, \chi, D) = w_D^h + a_D^h + \beta(1 - \mu) \mathbb{E}_{h'|h,D} \left[\delta^{h'} U(h', \chi') + (1 - \delta^{h'}) V(h', D, \chi') \right].$$

The presence of a continuation value makes the worker's decision a dynamic one. This added component implies that $\frac{\partial P_{D \rightarrow D}^h}{\partial w_D^h}$ is a forward-looking expression. To see this, note that

$$\begin{aligned} \frac{\partial v(h, D, \chi, D)}{\partial w_D^h} &= 1 + \beta(1 - \mu) \mathbb{E}_{h'|h,D} \left[\delta^{h'} \frac{\partial U(h', \chi')}{\partial w_D^h} + (1 - \delta^{h'}) \frac{\partial V(h', D, \chi')}{\partial w_D^h} \right] \\ &= 1 + \beta(1 - \mu) \left[\mathbb{1}_{h < h_H} g_E \left(\delta^{h+1} \frac{\partial U(h+1, \chi')}{\partial w_D^h} + (1 - \delta^{h+1}) \frac{\partial V(h+1, D, \chi')}{\partial w_D^h} \right) \right. \\ &\quad \left. + (1 - \mathbb{1}_{h < h_H} g_E) \left(\delta^h \frac{\partial U(h, \chi')}{\partial w_D^h} + (1 - \delta^h) \frac{\partial V(h, D, \chi')}{\partial w_D^h} \right) \right] \end{aligned}$$

where $U(h', \chi')$ is the value of being separated (given by Equation (4)), $V(h, D, \chi')$ is the worker value next period,

$$V(h, D, \chi') = \sigma \left(\ln \left(\exp \frac{v(h, D, \chi', D)}{\sigma} + M \exp \frac{v(h, D, \chi', P)}{\sigma} + \exp \frac{v(h, D, \chi', N)}{\sigma} \right) + \gamma^{EM} \right).$$

Again, because firms are non-atomistic, the choice of policies today affects the worker value next period through the change in the distribution $\chi' = (e'_D, n', s')$. Applying

the chain rule to the worker value like we did to the firm value above,

$$\frac{\partial V(h, D, \chi')}{\partial w_D^h} = \sum_{\hat{h}} \left[\frac{\partial V(h, D, \chi')}{\partial (e_D^{\hat{h}})'} \frac{\partial (e_D^{\hat{h}})'}{\partial w_D^h} + \frac{\partial V(h, D, \chi')}{\partial (s^{\hat{h}})'} \frac{\partial (s^{\hat{h}})'}{\partial w_D^h} + \frac{\partial V(h, D, \chi')}{\partial (n^{\hat{h}})'} \frac{\partial (n^{\hat{h}})'}{\partial w_D^h} \right].$$

Clearly, $\partial V(h, D, \chi')/\partial w_D^h$ is a function of $\partial v(h, D, \chi', D)/\partial w_D^h$, which is itself a function of $\partial V(h, D, \chi'')/\partial w_D^h$, and so on and so forth.

B.7 Non-concavity

I show by counterexample that the dominant firm's wage-setting objective need not be concave or quasi-concave, even when peripheral wages adjust optimally. I illustrate this by using the static limiting case of the model that most closely resembles the so-called “logit profit” models in the Operations Research literature (e.g. [Dong, Kouvelis, and Tian 2009](#), [Gallego and Wang 2014](#), and [Song, Xue, and Shen 2026](#)) which have long been known to be non-concave ([Hanson and Martin 1996](#)).

Lemma 1 (Concavity). *The dominant firm's objective in Equation (10) need not be concave or quasi-concave in the wage schedule w_D .*

Proof. By counterexample. Suppose $M = 1$ and there are two human capital levels, low (L , with $h = 1$) and high (H , with $h = 2$). Consider the following limiting parameterization, with zero weight on future payoffs, no switching costs, and no separations:

$$(\beta = 0, b^L = 23, b^H = 31, a_D^L = 0, a_D^H = 0, \phi^{EE} = \phi^{PD} = \phi^{DP} = \phi^{NE} = 0, \\ \delta^L = \delta^H = 0, \sigma = 1, p_D = 100).$$

Consider the following two wage schedules for the dominant firm

$$w_D^1 := (w_{D,L}^1, w_{D,H}^1) = (27, 0)$$

$$w_D^2 := (w_{D,L}^2, w_{D,H}^2) = (0, 34)$$

Using Equation (12), peripheral wages at these schedules, and at their convex combination below, are

$$w_P(w_D) := (w_P^L(w_D), w_P^H(w_D)) \approx (0, 1).$$

Since there are no switching costs, the conditional choice probabilities $P(d|h, o, \chi)$ do not depend on the origin o so I instead write $P(d|h)$ for their common value. The dominant firm's labor supply curve is then the product of the aggregate distribution of human capital and $P(D|h)$. Suppose half of the workers have low human capital. Writing $J_D(w_D)$ here for the dominant firm's objective at this fixed state gives

$$J_D(w_D) = \frac{1}{2}P(D|L)(p_D \cdot 1 - w_D^L) + \frac{1}{2}P(D|H)(p_D \cdot 2 - w_D^H),$$

where e.g. $P(D|h) = \frac{e^{w_D^h}}{e^{w_D^h} + e^{w_P^h} + e^{b^h}}$. For the two schedules

$$\begin{aligned} J_D(w_D^1) &\approx \frac{1}{2} \frac{e^{27}}{e^{27} + e^0 + e^{23}}(100 - 27) + \frac{1}{2} \frac{e^0}{e^0 + e^1 + e^{31}}(200 - 0) \approx 35.8 \\ J_D(w_D^2) &\approx \frac{1}{2} \frac{e^0}{e^0 + e^0 + e^{23}}(100 - 0) + \frac{1}{2} \frac{e^{34}}{e^{34} + e^1 + e^{31}}(200 - 34) \approx 79.1 \end{aligned}$$

Now consider the following convex combination of these wage schedules

$$w_D^3 := \frac{1}{2} \times w_D^1 + \frac{1}{2} \times w_D^2 = (13.5, 17).$$

This is below the value of non-employment at both L and H , attracting almost no workers of either type, and

$$\begin{aligned} J_D(w_D^3) &\approx \frac{1}{2} \frac{e^{13.5}}{e^{13.5} + e^0 + e^{23}}(100 - 13.5) + \frac{1}{2} \frac{e^{17}}{e^{17} + e^1 + e^{31}}(200 - 17) \approx 0.003 \\ &< \frac{1}{2} J_D(w_D^1) + \frac{1}{2} J_D(w_D^2) \approx 57.5 \end{aligned}$$

so the dominant firm's profits are not concave. We also have

$$J_D(w_D^3) \approx 0.003 < \min\{J_D(w_D^1), J_D(w_D^2)\} \approx 35.8$$

so the dominant firm's profits are not quasi-concave. Both inequalities are strict and the objective is continuous in β , so they also hold for $\beta > 0$ small enough. $\square$

B.8 Calibrating the Number of Peripheral Firms

The transition rates used in estimation pool workers across many markets with different numbers of firms. Suppose the markets in a representative group differ only

in their firm count. Let F be the worker-weighted distribution of these counts and $P(\theta_S, M) \in \mathbb{R}^{30}$ the model's transition rates at observed wages in a market with M peripheral firms. I approximate the pooled rates by the average $\int P(\theta_S, M) dF(M)$, which weights markets by employment.⁴¹ The calibrated M solves

$$M_{\text{eff}}(\theta_S) := \arg \min_M \sum_{j=1}^{30} \left| P_j(\theta_S, M) - \int P_j(\theta_S, M') dF(M') \right|. \quad (\text{B.19})$$

In practice, I use a log-normal fitted to the worker-weighted median and mean firm count of each representative market (bottom: median 634, mean 2,988; top: median 99, mean 260). I calibrate M separately from the other model parameters because it is only weakly identified separately from σ and ϕ^{EE} . This is because changes in the firm count can be partly offset by changes in the taste-shock scale and switching costs, leaving the transitions nearly unchanged. Because Equation (B.19) depends on θ_S , the calibration and the estimation form a fixed point: I compute M_{eff} at preliminary estimates, re-estimate θ_S at the resulting count, and iterate until the recomputed M_{eff} differs from the count used in estimation by less than 10%. In Table B.1, I re-estimate the supply parameters at other values of M and find that the fit is nearly flat and the estimated markdowns move by at most one percentage point.

Table B.1: Sensitivity of Supply Estimates to the Calibrated Firm Count

	Specification	M	CCP fit (MAE)	Lerner ratio η_P	Lerner ratio η_D
Bottom	Weighted median	634	0.0055	[0.69, 0.73, 0.74]	[0.86, 0.89, 0.90]
	Calibrated baseline	667	0.0055	[0.69, 0.73, 0.74]	[0.86, 0.89, 0.90]
	Weighted mean	2,988	0.0059	[0.70, 0.74, 0.75]	[0.86, 0.89, 0.91]
Top	Weighted median	99	0.0053	[0.66, 0.70, 0.72]	[0.75, 0.78, 0.80]
	Calibrated baseline	127	0.0052	[0.66, 0.70, 0.72]	[0.75, 0.79, 0.80]
	Weighted mean	260	0.0052	[0.66, 0.70, 0.72]	[0.75, 0.79, 0.80]

Note: Each row re-estimates the twelve supply parameters for the given value of M . Ratios reported as $[h_1, h_2, h_3]$.

⁴¹ Pooling workers implicitly weights each market by its number of workers in each human capital-origin group. I approximate these weights using total market employment.

B.9 Removing Learning and Turnover

I extend Section 5.3 by also removing learning and turnover, which holds the aggregate human capital distribution fixed.⁴² Table B.2 reports the results. The welfare losses fall to 0.50% in the bottom market and 0.72% in the top, understating the full model’s losses by 70% and 79%. This comparison combines the productivity mis-measurement with changes in skills and reallocation costs. In particular, removing turnover eliminates the replacement of non-employed workers by employed entrants, so employment must expand through workers’ costly moves into employment. The exercise therefore does not isolate the contribution of learning.

Table B.2: Welfare Comparisons With and Without Learning and Turnover

	Bottom			Top		
	Static p_D	No learning or turnover	Full Model	Static p_D	No learning or turnover	Full Model
Welfare gap (%)	−1.58	−0.50	−1.64	−2.08	−0.72	−3.47
Output gap (%)	−7.59	−7.33	−7.67	−8.95	−8.43	−9.96
Planner non-employment (%)	12.87	12.87	12.87	9.86	9.95	9.87
Planner dominant employment (%)	0.08	0.06	0.05	13.10	9.60	2.39
Planner mean human capital	4.490	4.475	4.490	4.118	4.108	4.118

Note: Static p_D and Full Model reproduce Table 10. No learning or turnover uses static p_D and removes learning and turnover from the efficient benchmark’s law of motion. Mean human capital is measured among employed workers, in units of 10,000€ of annual output at peripheral firms (where $p_P = 1$).

C Computational Appendix

C.1 Solution Method

I solve for the model’s equilibrium by fitted value function iteration, where the worker values $V(h, o, \chi)$ and $U(h, \chi)$ and the dominant firm’s value $J_D(\chi)$ are approximated by neural networks. The dominant firm’s wages are found by multi-start numerical optimization at each training state, with peripheral wages determined by the Lerner rule inside the within-period fixed point.

⁴² Worker discounting remains $\beta(1 - \mu)$ with $\mu = 0.25$. Turnover is removed only from the law of motion.

Function approximation. The worker network $\text{NN}_V : \Delta^{3H} \rightarrow \mathbb{R}^{4H}$ has parameters θ_V and outputs $(U(h, \chi), V(h, D, \chi), V(h, P, \chi), V(h, N, \chi))_{h=1}^H$. The firm network $\text{NN}_J : \Delta^{3H} \rightarrow \mathbb{R}$ has parameters θ_J and outputs $J_D(\chi)$. Both map the $3H$ state shares through a dense layer of width 128, two pre-norm residual blocks with leaky ReLU activations and layer normalization (He et al. 2016, Xiong et al. 2020), and a dense output layer. Weights are initialized as in He, Zhang, Ren, and Sun (2015) and trained with Adam (Kingma and Ba 2015) on a cosine learning-rate schedule.

The inner fixed point. Evaluating a candidate dominant firm wage involves solving a fixed point. Worker choices depend on continuation values at χ' , but those choices determine χ' through the law of motion. Peripheral wages also depend on worker choices through the labor supply elasticity. The joint system is

$$\chi' = \Gamma(\chi, P(w_D, w_P, V(\cdot, \cdot, \chi'))), \quad w_P = \eta_P(\chi, w_D, w_P, V(\cdot, \cdot, \chi')) \cdot h, \quad (\text{C.1})$$

where P denotes the choice probabilities and Γ the law of motion. The second equation is the peripheral Lerner rule (Equation (12)). The dependence of worker choices on the aggregate state they generate arises because the dominant firm is non-atomistic (Appendix B.6). Hence, for each state χ and candidate dominant-firm wage w_D , I jointly determine the next-period state χ' and peripheral wages w_P . I iterate on this system four times, starting from $\chi' = \chi$.⁴³ To construct the firm’s gradient targets, I instead solve the joint system to convergence and differentiate it implicitly, as described below.

Initialization and training states. Before training, I solve the worker Bellman equations at observed wages to obtain V^{data} and pre-train NN_V to return these values at every state. I initialize NN_J with random weights and set the wage policy to w_D^{data} . The training pool $\{\chi^{(m)}\}_{m=1}^{N_{\text{pool}}}$ is a mix between a fixed Sobol sequence, comprising 90% of the pool and including χ_{data} , and random draws centered around the current estimate of the stationary distribution, which is updated each outer iteration.

Because network parameters are shared across states, a training step at one state also changes the continuation values at others, so Bellman targets computed from the network being trained would move during fitting. To prevent this, each network has a

⁴³ In practice, I have found four iterations to yield a very good approximation to the fixed point.

target copy, NN_V^{tgt} and NN_J^{tgt} , held fixed within each outer iteration of the algorithm (Mnih et al., 2015). The copies are reset to the current networks at the end of each outer iteration.

Worker update. I draw a batch of training states $\mathcal{B}_V$. At each state $\chi^{(m)}$, I use the stored wage $\hat{w}_D^{(m)}$ and the inner fixed point to compute the right-hand sides of Equations (3) and (4), with continuation values given by the target network. I then fit NN_V to these Bellman targets by minimizing their mean-squared distance from the network’s output.

Firm update. I draw a second batch $\mathcal{B}_Q$ and solve the dominant firm’s wage problem at each state:

$$\hat{w}_D^{(m)} = \arg \max_{w_D \in \mathbb{R}_+^H} F_D(m_D^{(m)}) - \sum_h m_D^{h,(m)} w_D^h + \beta \text{NN}_J^{\text{tgt}}(\chi'^{(m)}(w_D)). \quad (\text{C.2})$$

I use n_{starts} starting wages because the firm’s value is non-concave (Lemma 1). Wages are parameterized as cumulative increments, $w_D^h = \sum_{j \leq h} z^j$ with $z^j \geq 0$, imposing a non-decreasing wage schedule.⁴⁴ The maximized value provides the value target $\hat{J}^{(m)}$ for NN_J .

Value-gradient training. Accurate value levels need not deliver accurate wage choices. A change in wages moves the next-period distribution, so the firm’s incentive to raise wages depends on the gradient of continuation value. Small errors in fitted values can leave this gradient poorly approximated which can change which wage policy gets chosen. I therefore fit the firm’s value and its gradient jointly (Czarnecki et al. 2017). The envelope theorem applied to Equation (C.2) gives the gradient Bellman equation at the optimum $\hat{w}_D$:

$$\nabla J_D(\chi) = \nabla_\chi \left(F_D(m_D) - \sum_h m_D^h \hat{w}_D^h \right) + \beta \left(\frac{d\chi'}{d\chi} \right)^\top \nabla J_D(\chi'). \quad (\text{C.3})$$

Both derivatives include the response of worker choices and peripheral wages through the joint fixed point. The dominant firm’s own wage is held fixed when taking these

⁴⁴ This is for speed: a search without this restriction finds the same maximum at the observed and stationary states.

derivatives, as its response drops out by the envelope theorem. I obtain the derivatives by implicitly differentiating the converged system in Equation (C.1), avoiding differentiation through the four-pass approximation. At each state in $\mathcal{B}_Q$, I form the gradient target $g^{(m)}$ from the right-hand side of Equation (C.3), using $\nabla \text{NN}_J^{\text{tgt}}(\chi'^{(m)})$ for the continuation gradient. I then minimize the loss function

$$\mathcal{L}_J = \frac{1}{|\mathcal{B}_Q|} \sum_m (\text{NN}_J(\chi^{(m)}) - \hat{J}^{(m)})^2 + \lambda \frac{1}{3H|\mathcal{B}_Q|} \sum_m \sum_{i=1}^{3H} \omega_i (\partial_i \text{NN}_J(\chi^{(m)}) - g_i^{(m)})^2, \quad (\text{C.4})$$

where ω_i are weights for scale normalization.

Implementation. The estimated solutions use 200 outer iterations and $N_{\text{pool}} = 10,000$ training states. Each iteration samples $|\mathcal{B}_V| = 2,048$ states for the worker update and $|\mathcal{B}_Q| = 1,024$ for the firm update. Each update takes 200 Adam steps, with a cosine learning-rate schedule from 2×10^{-3} to 2×10^{-4} . The wage search uses $n_{\text{starts}} = 6$ starts with at most 400 objective evaluations each, and the gradient-loss weight is $\lambda = 1$. I also dampen policy updates to help convergence.

C.2 Accuracy

I now assess accuracy of the solution method at the estimated parameter values using equilibrium residuals, profit-based wage checks, and a comparison with polynomial projection. Following Azinovic-Yang and Žemlička (2025), I express the equilibrium residuals in economic units and report their distribution over simulated states.⁴⁵

Simulated states. The results hinge on the accuracy of the solution method along transition paths, not just at the observed and stationary states. I therefore evaluate the solution on 21 equilibrium paths of 61 years, giving 1,281 states per market. One path starts at the observed state. The other 20 paths start from initial distributions chosen to assess accuracy over a wider range of market states. The six- and twenty-start wage searches reach the same maximum at every simulated state.

⁴⁵ The solution also reproduces closed forms where they exist. At the stationary state, where wages are constant, worker values match the closed-form solution to 0.08% and 0.05% of consumption and the wage equation (Equation (11)) holds to 0.008%.

Worker errors. I measure worker errors by the residual in their Bellman equations. Let $r(\chi)$ be the gap between $\text{NN}_V(\chi)$ and the right-hand side of Equations (3) and (4). Since utility is linear in consumption, this residual is the change in current consumption needed to reconcile the network value with its Bellman target. I report $|r|/\bar{w}^h$, where $\bar{w}^h$ is the mean wage at skill level h , taking the largest error across the twelve values at each state. Mean errors are 0.13% in the bottom market and 0.11% in the top, with 90th percentiles of 0.16% and 0.08% and 99.9th percentiles of 1.85% and 3.27% (Table C.1).

Firm errors. For the leader, I translate the error in its first-order condition into an implied wage error. The continuation value enters the wage choice only through $\beta (\partial \chi' / \partial w_D)^\top \nabla \text{NN}_J(\chi')$. Replacing the network gradient by its target $g(\chi')$, the right-hand side of Equation (C.3) at χ' , moves the optimal wage by

$$\Delta w_D(\chi) = [\nabla_{w_D}^2 \Phi(\hat{w}_D; \chi)]^{-1} \beta \left(\frac{\partial \chi'}{\partial w_D} \right)^\top (\nabla \text{NN}_J(\chi') - g(\chi')), \quad (\text{C.5})$$

where Φ is the objective in Equation (C.2). I report $|\Delta w_D^h|/\hat{w}_D^h$, taking the largest error across skill levels at each state. Mean wage errors are 0.24% in the bottom market and 0.08% in the top, with 90th percentiles of 0.26% and 0.07% and 99.9th percentiles of 0.51% and 0.31% (Table C.1). For comparison, the estimated model misses observed leader wages by 4.0% and 1.2% (Table 6). The leader's Bellman residual averages 0.072% and 0.006% of its value at the observed state. Across the simulated states, the network gradient fits its target with an R^2 of 0.984 and 0.9997. Relative to a 40-iteration calculation at the same states and dominant-firm wages, four iterations change peripheral wages by at most 0.026% in the bottom market and 0.0094% in the top.

Table C.1: Errors in equilibrium conditions

	Bottom			Top		
	Mean	p_{90}	$p_{99.9}$	Mean	p_{90}	$p_{99.9}$
Worker value (%)	0.127	0.159	1.850	0.109	0.084	3.273
Firm value (%)	0.072	0.072	0.180	0.006	0.006	0.020
Wage policy (%)	0.243	0.255	0.508	0.076	0.074	0.309

Note: Worker value and wage policy errors are maxima at each state.

Profit-based wage checks. At the observed and stationary states, I check one-period wage deviations by computing 300 years of discounted profits, using the network’s worker values and subsequent wage policy. This avoids using the firm’s value network. The implied optimal wage differs from the equilibrium wage by less than 0.3%, consistent with Equation (C.5).

Independent solution. I also solve the model by polynomial projection (Judd, 1992, Judd et al., 2011). I approximate the twelve worker values and the dominant firm’s value with degree-four polynomials fitted by least squares over a grid of distributions around the observed and stationary states. This restricted domain covers 99% of the simulated states, providing a local check on the network solution trained over a broader domain. Policy iteration alternates Bellman updates with wage searches from the previous and observed wages, using the same within-period fixed point as the network solution. I stop when the 99th percentile of wage changes falls below 0.03%. Table C.2 compares the two solutions: at every simulated state the leader’s wages differ by less than 1%.

Table C.2: Dominant wage comparison

Market	Skill	Observed state		Simulated states		
		Degree 3	Degree 4	Mean	p_{90}	Max
Bottom	h_1	−0.39	+0.15	+0.23	0.27	0.98
	h_2	+0.20	+0.77	+0.73	0.74	0.81
	h_3	−0.92	−0.57	−0.68	0.68	0.99
Top	h_1	−0.40	−0.42	−0.47	0.47	0.77
	h_2	−0.36	−0.33	−0.25	0.25	0.60
	h_3	−0.12	−0.09	−0.15	0.15	0.59

Note: Differences between projection and network wages, in % of the network wage. Simulated-state comparisons use degree-four polynomials. p_{90} and maximum use absolute differences.

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